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Measure Theory

1.1 Outer Measure

Definition 1.1.1 ▶ Outer Measure

A function φ defined for every subset A of an arbitrary set X is called an ***outer measure*** on X if the following conditions are satisfied:

(i) $\varphi(\varnothing) = 0$,

(ii) $0 \leq \varphi(A) \leq \infty$ whenever $A \subset X$,

(iii) $\varphi(A_1) \leq \varphi(A_2)$ whenever $A_1 \subset A_2$,

(iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for every countable collection of sets $\{A_i\}$ in X.

Example 1.1.2 ▶ Outer Measure

(i) In an arbitrary set X, define $\varphi(A) = 1$ if A is nonempty and $\varphi(\varnothing) = 0$.

(ii) Let $\varphi(A)$ be the number (possibly infinite) of points in A.

(iii) Let $\varphi(A) = \begin{cases} 0 & \text{if card } A \leq \aleph_0, \\ 1 & \text{if card } A > \aleph_0. \end{cases}$

(iv) If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A.

(v) Select a fixed x_0 in an arbitrary set X, and let

$$\varphi(A) = \begin{cases} 0 & \text{if } x_0 \notin A, \\ 1 & \text{if } x_0 \in A; \end{cases}$$

φ is called the ***Dirac measure*** concentrated at x_0.

Definition 1.1.3

Let φ be an outer measure on a set X. A set $E \subset X$ is called φ***-measurable*** if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A - E)$$

for every set $A \subset X$. In view of property (iv) above, observe that φ-measurability requires only

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E).$$

Note.
也就是说, E 是可测的, 当且仅当任意集合 A 都可以被 E 按外测度被良好地拆解为 $A \cap E$ 和 $A \setminus E$.

Lemma 1.1.4

A set $E \subset X$ is φ-measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q)$$

for all sets P and Q such that $P \subset E$ and $Q \subset E^c$.

Note.
也就是说 E 可测, 当且仅当落在 E 的内, 外的集合在外测度意义下是无关的.

Proof. For Necessity, if E is φ-measurable , then for all such P, Q,

$$\varphi\left(P \cup Q\right) = \varphi\left(\left(P \cup Q\right) \cap E\right) + \varphi\left(\left(P \cup Q\right) - E\right) = \varphi\left(P\right) + \varphi\left(Q\right)$$

For sufficiency , let $P = A \cap E$, $Q = A \cap E^c = A \setminus E$. Then

$$\varphi\left(A\right) = \varphi\left(P \cup Q\right) = \varphi\left(P\right) + \varphi\left(Q\right) = \varphi\left(A \cap E\right) + \varphi\left(A \setminus E\right)$$

□

Theorem 1.1.5

Suppose φ is an outer measure on an arbitrary set X. Then the following four statements hold:

(i) E is φ-measurable whenever $\varphi(E) = 0$.
(ii) $\varnothing$ and X are φ-measurable.
(iii) $E_1 - E_2$ is φ-measurable whenever E_1 and E_2 are φ-measurable.
(iv) If $\{E_i\}$ is a ***countable collection of disjoint φ-measurable sets***, then

$\bigcup_{i=1}^{\infty} E_i$ is φ-measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

More generally, if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c),$$

where

$$S = \bigcup_{i=1}^{\infty} E_i.$$

Proof. 1. Take any $A \subseteq X$. Since $A \cap E \subseteq E$, $\varphi(A \cap E) = 0$. Thus

$$\varphi(A \cap E) + \varphi(A \cap E^c) = \varphi(A \cap E^c) \leq \varphi(A)$$

2. Follows immediately frome lemma 1.1.4.

3. Take any P, Q such that $P \subseteq (E_1 \setminus E_2) = E_1 \cap E_2^c$, $Q \subseteq (E_1 \setminus E_2)^c = E_1^c \cup E_2$.

$$\varphi(Q) = \varphi(Q \cap E_2) + \varphi(Q - E_2)$$

Note that $Q - E_2 \subseteq E_1^c$, $Q \cap E_2 \subseteq E_2$. Since $P \subseteq E_1$, the measurability of E_1 implis that

$$\varphi(P) + \varphi(Q - E_2) = \varphi(P \cup (Q - E_2))$$

Note that $P \cup (Q - E_2) \subseteq E_2^c$. Since $Q \cap E_2 \subseteq E_2$, we have

$$\varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) = \varphi(P \cup Q)$$

Finally , we have

$$\begin{aligned}\varphi(P) + \varphi(Q) &= \varphi(P) + \varphi(Q \cap E_2) + \varphi(Q - E_2)\\ &= \varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2)\\ &= \varphi(P \cup Q)\end{aligned}$$

4. Let $S_k = \bigcup_{i=1}^{k} E_i$. We proceed by finite induction and first note that the result is obviously true for $k = 1$. For $k > 1$ assume that S_k is φ-measurable and that

$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right), \quad \forall A \subseteq X$$

Then

$$\begin{aligned}\varphi(A) &= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap E_{k+1}^c\right) \\ &= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap E_{k+1}^c \cap S_k^c\right) + \varphi\left(A \cap E_{k+1}^c \cap S_k\right) \\ &= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right) + \varphi(A \cap S_k)\end{aligned}$$

Replace A by $A \cap S_k$, we have

$$\varphi(A \cap S_k) \geq \sum_{i=1}^{k} \varphi(A \cap S_k \cap E_i) + \varphi\left(A \cap S_k \cap S_k^c\right) = \sum_{i=1}^{k} \varphi(A \cap E_i)$$

Thus

$$\varphi(A) \geq \sum_{i=1}^{k+1} \varphi(A \cap E_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right)$$

Then the subadditivity gives that

$$\varphi(A) \geq \varphi(A \cap S_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right)$$

this , in turn , implies that S_{k+1} is φ-measurable. Since we now know for every set $A \subseteq X$ and for all positive integers k

$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right)$$

We have

$$\varphi(A) \geq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c) \geq \varphi(A \cap S) + \varphi(A \cap S^c)$$

where $S = \bigcup_{i=1}^{\infty} E_k$. Thus S is measurable. The subadditivity gives that

$$\varphi(A) \leq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Finally, we have

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Replace A by S , we can get

$$\varphi(S) = \sum_{i=1}^{\infty} \varphi(E_i)$$

$\square$

Lemma 1.1.6

Let $\{E_i\}$ be a sequence of arbitrary sets. Then there exists a sequence of disjoint sets $\{A_i\}$ such that each $A_i \subset E_i$ and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ-measurable, and so is A_i.

Proof. Let $S_k = \bigcup_{i=1}^{k} E_i$. Let $A_k = S_k \setminus S_{k-1}, k \geq 1$, where $S_0 = \varnothing$. Then

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} (S_k \setminus S_{k-1}) = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ-measurable. $E_{i+1} \cup E_i = (E_{i+1} - E_i) \cup E_i$ is measurable follows from the (ii), (iii) of the above theorem . Inductively , every $S_k = (S_{k-1} \setminus E_k) \cup E_k$ is measurable. Then each A_k is measurable . $\square$

Theorem 1.1.7

If $\{E_i\}$ is a sequence of φ-***measurable*** sets in X, then $\cup_{i=1}^{\infty} E_i$ and $\cap_{i=1}^{\infty} E_i$ are φ-***measurable***.

Proof. By the lemma above, there exists a sequence of disjoint φ - measurable sets $\{A_i\}$ such that each $A_i \subseteq E_i$, and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i$$

It follows from Theorem 1.1.5 that $\bigcup_{i=1}^{\infty} A_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i$ is as well . Since X is measurable , $E_i^c = X \setminus E_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i^c$ is measurable . Thus

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \bigcup_{i=1}^{\infty} E_i^c$$

is measurable . □

Definition 1.1.8

A nonempty collection Σ of sets E satisfying the following two conditions is called a σ-***algebra***:

(i) if $E \in \Sigma$, then $E^c \in \Sigma$;

(ii) $\cup_{i=1}^{\infty} E_i \in \Sigma$ if each E_i is in Σ.

Definition 1.1.9

In a topological space, the elements of the smallest σ-algebra that contains all open sets are called ***Borel sets***. The term "smallest" is taken in the sense of inclusion.

Proof. We need to show that the "smallest" exists. □

Corollary 1.1.10

If φ is an ***outer measure*** on an arbitrary set X, then the class of φ-measurable sets forms a σ-***algebra***.

Proof. Follows immediately from Theorem 1.1.5 and Theorem 1.1.7. □

Corollary 1.1.11

Suppose φ is an outer measure on X and $\{E_i\}$ a countable collection of φ-measurable sets.

(i) If $E_1 \subset E_2$ with $\varphi(E_1) < \infty$, then

$$\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1).$$

(ii) (Countable additivity) If $\{E_i\}$ is a disjoint sequence of sets, then

$$\varphi(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \varphi(E_i).$$

(iii) (Continuity from the left) If $\{E_i\}$ is an increasing sequence of sets, that is, if $E_i \subset E_{i+1}$ for each i, then

$$\varphi(\bigcup_{i=1}^{\infty} E_i) = \varphi(\lim_{i\to\infty} E_i) = \lim_{i\to\infty} \varphi(E_i).$$

(iv) (Continuity from the right) If $\{E_i\}$ is a decreasing sequence of sets, that is, if $E_i \supset E_{i+1}$ for each i, and if $\varphi(E_{i_0}) < \infty$ for some i_0, then

$$\varphi(\bigcap_{i=1}^{\infty} E_i) = \varphi(\lim_{i\to\infty} E_i) = \lim_{i\to\infty} \varphi(E_i).$$

(v) If $\{E_i\}$ is any sequence of φ-measurable sets, then

$$\varphi(\liminf_{i\to\infty} E_i) \leq \liminf_{i\to\infty} \varphi(E_i).$$

(vi) If

$$\varphi(\bigcup_{i=i_0}^{\infty} E_i) < \infty$$

for some positive integer i_0, then

$$\varphi(\limsup_{i\to\infty} E_i) \geq \limsup_{i\to\infty} \varphi(E_i).$$

Proof. (v) Let $A_j = \bigcap_{k=j}^{\infty} E_k$, then

$$\liminf_{i\to\infty} E_i = \lim_{j\to\infty} A_j = \bigcup_{j=1}^{\infty} A_j$$

It follows from (iv) that

$$\varphi\left(\liminf_{i\to\infty} E_i\right) = \lim_{j\to\infty} \varphi\left(A_j\right)$$

Since $A_j \subseteq E_j$, we have $\varphi\left(A_j\right) \leq \varphi\left(E_j\right)$, then

$$\lim_{j\to\infty} \varphi\left(A_j\right) \leq \liminf_{i\to\infty} \varphi\left(E_j\right)$$

Thus

$$\varphi\left(\liminf_{i\to\infty} E_i\right) \leq \liminf_{i\to\infty} \varphi\left(E_i\right)$$

□

Definition 1.1.12

- An outer measure φ on a topological space X is called a ***Borel outer measure*** if all Borel sets are φ-measurable.
- A Borel outer measure is ***finite*** if $\varphi(X)$ is finite.
- An outer measure φ on a set X is called ***regular*** if for each $A \subseteq X$ there exists a φ-measurable set $B \supseteq A$ such that $\varphi(B) = \varphi(A)$.
- A ***Borel rugular outer measure*** is a Borel outer measure such that for each $A \subseteq X$, there exists a Borel set B such that $\varphi(B) = \varphi(A)$.
- A ***Radon outer measure*** is a Borel regular outer measure that is finite on compact sets.

Exercises for Outer Measure

Exercise 1.1.13

If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A.

Let $\varphi_\varepsilon(A) := \varphi(A)$ denote the dependence on ε, and define

$$\psi(A) := \lim_{\varepsilon \to 0} \varphi_\varepsilon(A)$$

What is $\psi(A)$ and what are the correspoding ψ-measurable sets.

Proof. If there are only finite points in A. Let

$$l := \min\{d(x,y) : x, y \in A, x \neq y\}$$

Then for any $x, y \in A$, $x \neq y$. If B_1, B_2 are ball of radius $\frac{l}{2}$ such that $x \in B_1, y \in B_2$. Then $B_1 \cap B_2 = \varnothing$. But each point in A must covered by one of such ball, which shows that $\psi(A) \geq |A|$. Since it is obvious that $\psi(A) \leq |A|$, then $\psi(A) = |A|$.

Now suppose that A is a infinite set. If $\psi(A) < \infty$, say $\psi(A) = n$. Then for any sufficiently small ε, there are balls $B_1, \cdots, B_n$ with radius ε cover A. Take any $n+1$ different points $p_1, \cdots, p_{n+1}$ of A. Let $\varepsilon = \frac{1}{2}\min\left\{d\left(p_i, p_j\right) : i \neq j\right\}$, then $B_1, \cdots, B_n$ cannot cover these $n+1$ points, thus cannot cover A, which is a contradiction. Thus we have $\psi(A) = \infty$.

We find that ψ is just the counting measure on X.Take any $E \subseteq X$,and take any $A \subseteq X$, if A is finite, then $A \cap E, A - E$ are as well, and $|A| = |A \cap E| + |A - E|$, that is $\psi(A) = \psi(A \cap E) + \psi(A - E)$. If A is infinite, then one of the $A \cap E$ and $A - E$ is as well. Then

$$\psi(A) = \infty = \psi(A \cap E) + \psi(A - E)$$

The above shows that E is ψ-measurable. Thus all subset of X are ψ-measurable.

□

Exercise 1.1.14

It was shown that $\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1)$ if $E_1 \subseteq E_2$ are φ-measurable with $\varphi(E_1) < \infty$. Prove that this result remains true if E_2 is not assumed to be φ-measurable.

Proof. Since E_1 is measurable,

$$\varphi(E_2) = \varphi(E_2 \cap E_1) + \varphi(E_2 - E_1)$$

but

$$\varphi(E_2 \cap E_1) = \varphi(E_1)$$

which completes the proof. □

1.2 Carathéodory Outer Measure

Definition 1.2.1 ▶ Carathéodory outer measure

An outer measure φ defined on a metric space (X, p) is called a ***Carathéodory outer measure*** if

$$\varphi(A \cup B) = \varphi(A) + \varphi(B)$$

whenever A, B are arbitrary subsets of X with $d(A, B) > 0$. Where

$$d(A, B) := \inf\{\rho(a, b) : a \in A, b \in B\}$$

Theorem 1.2.2

If φ is a Carathéodory outer measure on a metric space X, then all closed sets are φ-measurable.

Proof sketch.
用一列集合近似代替 C 是标准的想法. 证明的核心要素有: 1.C 是闭集, 使得按距离向外逼近的行为是良好的. 2. 距离这个属性与 Carathódory 条件是相适应的. 此外, 将命题转化为极限

$$\lim_{i \to \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

是标准的流程. 要证明该极限成立, 则需要一定的技巧. 这里将 $A \setminus C_i$ 和 $A \setminus C$ 相差的中间部分进行细分量化, 将原极限转化为

$$\lim_{i \to \infty} \sum_{j=i}^{\infty} \varphi(T_j)$$

的极限问题, 从而转化为级数的收敛性问题. 将集合隔一个分组, 变成两组两两相隔正距离的集合, 以便套用 Carathéodory 条件.

Proof. Let $C \subseteq X$ be any closed sets. Take $A \subseteq X$, only need to show that

$$\varphi(A) \geq \varphi(A \cap C) + \varphi(A \setminus C)$$

We assume that $\varphi(A) < \infty$. Define

$$C_i := \left\{x : d(x, C) \leq \frac{1}{i}\right\}$$

Since $d(A \setminus C_i, A \cap C) > 0$,

$$\varphi(A) \geq \varphi((A \setminus C_i) \cup (A \cap C)) = \varphi(A \setminus C_i) + \varphi(A \cap C)$$

Only need to show that

$$\lim_{i \to \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

Define

$$T_i := \left\{x \in A : \frac{1}{i+1} < d(x, C) \leq \frac{1}{i}\right\}$$

, then

$$A \setminus C \subseteq (A \setminus C_i) \cup \bigcup_{j=i}^{\infty} T_i$$

Thus

$$\varphi(A \setminus C) \leq \varphi(A \setminus C_i) + \sum_{j=i}^{\infty} \varphi(T_i)$$

Once we have $\sum_{j=1}^{\infty} \varphi(T_i) < \infty$, then $\lim_{i \to \infty} \sum_{j=i}^{\infty} \varphi(T_i) = 0$, the conclusion follows from it. To show this, note that $\mathrm{d}\left(T_i, T_j\right) > 0$ whenever $|i - j| \geq 2$. Then it by induction follows from Carathódory condition that

$$\sum_{j=1}^{\infty} \varphi(T_{2i}) \leq \varphi(A) < \infty$$

$$\sum_{j=1}^{\infty} \varphi\left(T_{2j-1}\right) \leq \varphi(A) < \infty$$

Thus $\sum_{j=1}^{\infty} \varphi(T_i) < \infty$, which completes the proof. □

Theorem 1.2.3

Suppose $\mathcal{F}$ is a family of subsets of a topological space X that contains all open and all closed subsets of X. Suppose also that $\mathcal{F}$ is closed under countable unions and countable intersections. Then $\mathcal{F}$ contains all Borel sets; that is, $\mathcal{B} \subset \mathcal{F}$.

Note.
可以将包含 $\mathcal{B}$ 描述成对以下三种集合的封闭性: 1. 开集 2. 补集 3. 可数并.
如果我们事先没有对补集的封闭性, 可以通过加强 1.3. 的条件, 使得当我们强制使得补集封闭后, 得到的新集族依旧满足 1.3. 具体地, 我们希望原集族对于开集的补集, 也就是闭集也封闭; 对于可数并的补集, 也就是可数交也封闭.

Proof. Let

$$\mathcal{H} = \mathcal{F} \cap \{A : A^c \in \mathcal{F}\}$$

Obseve that $\mathcal{H}$ contains all closed sets. Moreover, it is easily seen that $\mathcal{H}$ is closed under complementation and countable unions. Thus $\mathcal{H}$ is a σ-algebra that contains the open sets and therefore contains all Borel sets. □

Theorem 1.2.4

If φ is a Carathéodory outer measure on a metric space X, then the Borel sets of X are φ-measurable.

Proof. Since the collection of φ-measurable sets is itself a σ-algebra , it is closed under complements and countable union and countable intersection. Theorem 1.2.2 gives that φ-measurable sets contains closed sets, thus contains open sets as well. Then the conclusion follows from Theorem 1.2.3 . □

Exercises for this section

Exercise 1.2.5 ▶ Smallest σ-algebra containing open sets

Prove that in every topological space X, there exists a smallest σ-algebra that contains all open sets in X. That is, prove that there is a σ-algebra Σ that contains all open sets and has the property that if Σ_1 is another σ-algebra containing all open sets, then $\Sigma \subset \Sigma_1$. In particular, for $X = \mathbb{R}^n$, note that there is a smallest σ-algebra that contains all the closed sets in $\mathbb{R}^n$.

Proof. Let

$$\Sigma := \bigcap_{\mathcal{M}_\alpha \text{ is a } \sigma \text{ algebra containing } X} \mathcal{M}_\alpha$$

Then it is obvious that Σ is closed under complement, countable union, and contains all open sets. Thus Σ is a σ-algebra containing all open sets. □

Exercise 1.2.6

In a topological space X the family of Borel sets, $\mathcal{B}$, is by definition the σ-algebra generated by the closed sets. The method below is another way of describing the Borel sets using transfinite induction. You are to fill in the necessary steps:

1. For an arbitrary family $\mathcal{F}$ of sets, let

$$\mathcal{F}^* = \left\{ \bigcup_{k=1}^{\infty} E_k \;:\; \text{where either } E_i \in \mathcal{F} \text{ or } \tilde{E}_i \in \mathcal{F} \text{ for all } i \in \mathbb{N} \right\}.$$

   Let Ω denote the smallest uncountable ordinal. We will use transfinite induction to define a family $\mathcal{E}_\alpha$ for each $\alpha < \Omega$.

2. Let $\mathcal{E}_0$:= all closed sets := $\mathcal{K}$. Now choose $\alpha < \Omega$ and assume that $\mathcal{E}_\beta$ has been defined for each β such that $0 \le \beta < \alpha$. Define

$$\mathcal{E}_\alpha := \left(\bigcup_{0 \le \beta < \alpha} \mathcal{E}_\beta \right)^*$$

   and define

$$\mathcal{A} := \bigcup_{0 \le \alpha < \Omega} \mathcal{E}_\alpha.$$

3. Show that each $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. Show that $\mathcal{A} \subset \mathcal{B}$.
5. Now show that $\mathcal{A}$ is a σ-algebra to conclude that $\mathcal{A} = \mathcal{B}$.
   (a) Show that $\emptyset, X \in \mathcal{A}$.
   (b) Let $A \in \mathcal{A} \implies A \in \mathcal{E}_\alpha$ for some $\alpha < \Omega$. Show that this implies $\tilde{A} \in \mathcal{E}_\alpha^* \subset \mathcal{E}_\beta$ for every $\beta > \alpha$.
   (c) Conclude that $\tilde{A} \in \mathcal{A}$ and thus conclude that $\mathcal{A}$ is closed under complementation.
   (d) Now let $\{A_k\}$ be a sequence in $\mathcal{A}$. Show that $\bigcup_{k=1}^\infty A_k \in \mathcal{A}$. (Hint: Each A_k is in $\mathcal{A}_{\alpha_k}$ for some $\alpha_k < \Omega$. We know that there is $\beta < \Omega$ such that $\beta > \alpha_k$ for each $k \in \mathbb{N}$.)

Proof. 3. Since $\mathcal{E}_0 \subseteq \mathcal{B}$, only need to show that if $\mathcal{E}_\beta \subseteq \mathcal{B}$ for all $0 \leq \beta < \alpha$, then $\mathcal{E}_\alpha \subseteq \mathcal{B}$. Take any $E \in \mathcal{E}_\alpha$, then there are $E_1, \cdots, E_k, \cdots$, such that $E_i \in \bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta$, or $E_i^c \in \bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta$, and $E = \bigcup_{k=1}^\infty E_k$. We suppose that $E_i \in \mathcal{E}_{\beta_i}$ or $E_i^c \in \mathcal{E}_{\beta_i}$. Since $\mathcal{E}_{\beta_i} \subseteq \mathcal{B}$ for all β_i, and since $\mathcal{B}$ is closed under complement, then $E_i \in \mathcal{B}$. Since $\mathcal{B}$ is closed under countable union, $E = \bigcup_{k=1}^\infty E_k \in \mathcal{B}$. Thus $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. We have shown that $\mathcal{E}_\alpha \subseteq \mathcal{B}$ for all $\alpha < \Omega$, then it is immediate that $A \subseteq \mathcal{B}$.

5. (a) Since $\emptyset \in \mathcal{E}_0$ and $X = (\emptyset)^c \in \mathcal{E}_1 = (\mathcal{E}_0)^*$, then $\emptyset, X \in \mathcal{A}$.

(b) Write

$$\mathcal{E}_\alpha^* = \left\{ \bigcup_{k=1}^\infty E_k \ : \ \text{where either } E_i \in \mathcal{E}_\alpha \text{ or } E_i^c \in \mathcal{E}_\alpha \text{ for all } i \in \mathbb{N} \right\}$$

We take $E_1 = A^c$, then $E_1^c = A \in \mathcal{E}_\alpha$, and take $E_2 = \cdots = \emptyset$.Then $A^c = \bigcup_{k=1}^\infty E_k \in \mathcal{E}_\alpha^*$. If $\beta > \alpha$, then

$$\mathcal{E}_\alpha \subseteq \bigcup_{0 \leq \gamma < \beta} \mathcal{E}_\gamma$$

Let $*$ act on both side, then it follows from the definition of $*$ that

$$(\mathcal{E}_\alpha)^* \subseteq \left(\bigcup_{0 \leq \gamma < \beta} \mathcal{E}_\gamma \right)^* = \mathcal{E}_\beta$$

Thus $A^c \in \mathcal{E}_\alpha^* \subseteq \mathcal{E}_\beta$ for every $\beta > \alpha$.

(c) It is immediate from (b) that $A^c \in \mathcal{E}_\beta \subseteq \mathcal{A}$.

(d) Suppose that $A_k \in \mathcal{E}_{\alpha_k}$ for some $\alpha_k < \Omega$. Let

$$\gamma := \bigcup_{k\in\mathbb{N}} \alpha_k, \quad \beta := \gamma + 1$$

Then $\gamma, \beta < \Omega$, and $\alpha_1, \cdots, \alpha_k < \beta$. Then

$$\bigcup_{k=1}^{\infty} A_k \in \left(\bigcup_{k=1}^{\infty} \mathcal{E}_{\alpha_k}\right)^* \subseteq \mathcal{E}_\beta \subseteq \mathcal{A}$$

□

Exercise 1.2.7 ▶ *

An outer measure φ on a space X is called σ-finite if there exists a countable number of sets A_i with $\varphi(A_i) < \infty$ such that $X \subseteq \bigcup_{i=1}^\infty A_i$. Assuming that φ is a σ-finite Borel regular outer measure on a metric space X, prove that $E \subseteq X$ is φ-measurable if and only if there exists an F_σ set $F \subseteq E$ such that $\varphi(E \setminus F) = 0$.

Proof. Let $E_i := E \cap A_i$. Then

$$E = \bigcup_{i=1}^{\infty} E_i, \quad \varphi(E_i) \leq \varphi(A_i) < \infty$$

Once we show that there is a closed set $F_i \subseteq E_i$, such that $\varphi(E_i \setminus F_i) = 0$ for all i, then let $F = \bigcup_{i=1}^\infty F_i \in F_\sigma$, it follows that

$$\varphi(E \setminus F) \leq \varphi\left(\bigcup (E \setminus F_i)\right) \leq \varphi\left(\bigcup (E_i \setminus F_i)\right) \leq \sum_i \varphi(E_i \setminus F_i) = 0$$

□

Exercise 1.2.8

Let φ be an outer measure on a space X. Suppose $A \subset X$ is an arbitrary set with $\varphi(A) < \infty$ such that there exists a φ-measurable set $E \supset A$ with $\varphi(E) = \varphi(A)$. Prove that $\varphi(A \cap B) = \varphi(E \cap B)$ for every φ-measurable set

B.

Proof.

$$\varphi(A) = \varphi(E) = \varphi(E \cap B) + \varphi(E \setminus B)$$

$$\begin{aligned} \varphi(A \cap B) \geq \varphi(A) - \varphi(A \setminus B) &= \varphi(E) - \varphi(A \setminus B) \\ &= \varphi(E \cap B) + \varphi(E \setminus B) - \varphi(A \setminus B) \\ &\geq \varphi(E \cap B) \end{aligned}$$

Since $A \cap B \subseteq E \cap B$, $\varphi(A \cap B) \leq \varphi(E \cap B)$. Finally, we have $\varphi(A \cap B) = \varphi(E \cap B)$ ☐

Exercise 1.2.9

In $\mathbb{R}^2$, find two disjoint closed sets A and B such that $\mathbf{d}(A, B) = 0$. Show that this is not possible if one of the sets is compact.

Proof.

1. Let A be the graph of $y = \frac{1}{x}$. Let $B = \{(x, y) : y = 0\}$. Then $\left\{\left(n, \frac{1}{n}\right) : n \in \mathbb{N}\right\} \subseteq A$, $\{(n, 0) : n \in \mathbb{N}\} \subseteq B$, then
$$\mathbf{d}(A, B) \leq \inf\left\{d\left(\left(n, \frac{1}{n}\right), (n, 0)\right) : n \in \mathbb{N}\right\} = \lim_{n \to \infty} \frac{1}{n} = 0$$
2. If A is compact, B is closed. Since $\mathbf{d}(A, B) = 0$, there are convergent sequences $\{a_n\} \subseteq A$ and $\{b_n\} \subseteq B$ such that $\lim_{n\to\infty} d(a_n, b_n) = 0$. Since A is bounded and closed, there is a cluster point $a \in A$ of $\{a_n\}$. Suppose that $\lim_{k\to\infty} a_{n_k} = a$, then $\lim_{k\to\infty} b_{n_k} = a \in B$. $A \cap B \neq \emptyset$, which is a contradiction.

☐

Exercise 1.2.10

Let φ be an outer Carathéodory measure on $\mathbb{R}$ and let $f(x) := \varphi(I_x)$, where I_x is an open interval of fixed length centered at x. Prove that f is lower semicontinuous. What can you say about f if I_x is taken as a closed interval? Prove the analogous result in $\mathbb{R}^n$; that is, let $f(x) := \varphi(B(x, a))$, where $B(x, a)$ is the open ball with fixed radius a centered at x.

Exercise 1.2.11

In a metric space X, prove that $\operatorname{dist}(A,B) = d(\bar{A},\bar{B})$ for arbitrary sets $A, B \in X$.

Exercise 1.2.12

Let A be a non-Borel subset of $\mathbb{R}^n$ and define for each subset E,

$$\varphi(E) = \begin{cases} 0 & \text{if } E \subset A, \\ \infty & \text{if } E \setminus A \neq \varnothing. \end{cases}$$

Prove that φ is an outer measure that is not Borel regular.

Exercise 1.2.13

Let $\mathcal{M}$ denote the class of φ-measurable sets of an outer measure φ defined on a set X. If $\varphi(X) < \infty$, prove that the family

$$\mathcal{F} := \{A \in \mathcal{M} : \varphi(A) > 0\}$$

is at most countable.

1.3 Lebesgue Measure

Theorem 1.3.1

Suppose each edge $I_k = [a_k, b_k]$ of an n-dimensional interval I is partitioned into α_k subintervals. The products of these intervals produce a partition of I into $\beta := \alpha_1 \cdot \alpha_2 \cdots \alpha_n$ subintervals I_i and

$$v(I) = \sum_{i=1}^{\beta} v(I_i).$$

Theorem 1.3.2

For each interval I and each $\epsilon > 0$, there exists an interval J whose interior contains I and

$$v(J) < v(I) + \epsilon.$$

Definition 1.3.3 ▸ Lebesgue outer measure

The ***Lebesgue outer measure*** of an arbitrary set $E \subseteq \mathbb{R}^n$, denoted by $\lambda^*(E)$, is defined by

$$\lambda^*(E) := \inf\left\{\sum_{k=1}^{\infty} v(I_k)\right\},$$

where the infimum is taken over all countable collections of closed intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k.$$

Remark.

- It may be necessary at times to emphasize the dimension of the Euclidean space in which Lebesgue outer measure if defined. When clarification is needed, we will write $\lambda_n^*(E)$ in plce place of $\lambda^*(E)$.
- We have not prove that ***Lebesgue outer measure*** is actually a outer measure till now.

Theorem 1.3.4

For a closed interval $I \subseteq \mathbb{R}^n$, $\lambda^*(I) = v(I)$.

Proof. The inequality $\lambda^*(I) \le v(I)$ holds, since I itself givs a element of $\left\{\sum_{k=1}^{\infty} v(I_k)\right\}$ in the definition of $\lambda^*(I)$.

To prove the opposite inequality, choose $\varepsilon > 0$, and a sequence of closed intervals $I_1, I_2, \cdots$ such that

$$I \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) \le \lambda^*(E) + \varepsilon$$

We can choose closed intervals J_k such that

$$v(J_k) \le v(I_k) + \frac{\varepsilon}{2^k}, \quad I_k \subseteq J_k^\circ$$

Let $\mathcal{F} := \left\{J_k^\circ : k \in \mathbb{N}\right\}$. Then $\mathcal{F}$ is an open cover of I, since I is compact, it has a lebesgue number η. There is a finitely many partition $K_1, \cdots, K_m$ of I such that each with diameter less than η. Then K_i is contained in some element of $\mathcal{F}$, we say J_{k_i} althoung two K_i may be contained in a same J_{k_i}. Let N_m be the smallest number of J_{k_i} containing K_i, then

$$v(I) = \sum_{i=1}^{m} v(K_i) \leq \sum_{i=1}^{N_m} v\left(J_{k_i}\right) \leq \sum_{k=1}^{\infty} v(J_k) \leq \lambda^*(E) + 2\varepsilon$$

Since ε is arbitrary, $v(I) \leq \lambda^*(E)$. □

Theorem 1.3.5

Lebesgue outer measure, λ^*, defined on $\mathbb{R}^n$ is a Carathéorody outer measure.

Proof. We first verify that λ^* is an outer measure. We only show the subadditivity of λ^*, the other conditions are obvious. Let $\{A_i\}$ be a countable collection of arbitrary subsets of $\mathbb{R}^n$. Denote $A := \bigcup_{i=1}^{\infty} A_i$. There is a countable family of closed intervals $\left\{I_j^{(i)}\right\}_{j\in\mathbb{N}}$ such that

$$A_i \subseteq \bigcup_{j=1}^{\infty} I_j^{(i)}, \quad \sum_{j=1}^{\infty} v\left(I_j^{(i)}\right) < \lambda^*(A_i) + \frac{\varepsilon}{2^i}$$

Collection of closed interval $\left\{I_j^{(i)}\right\}_{i,j\in\mathbb{N}}$ can be aranged to a sequence of closed invervals, such that

$$A \subseteq \bigcup_{i,j\in\mathbb{N}} I_j^{(i)}$$

Thus

$$\lambda^*(A) \leq \sum_{i,j} v\left(I_j^{(i)}\right) \leq \sum_{i=1}^{\infty} \lambda^*(A_i) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, the countable subadditivity of λ^* is established.

To show that λ^* is Carathéorody outer measure. We take any $\varepsilon > 0$, and take a countable conllection of closed intervals $I_1, I_2, \cdots$ such that

$$A \cup B \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) \subseteq \lambda^*(A \cup B).$$

We can subdividing each I_k into smaller invervals if necessary, we may assume that each interval I_k which diameter less that $\mathbf{d}(A, B)$. Then $\{I_k\}$ can be seperated into two subfamiles $\{I'_k\}$ and $\{I''_k\}$. The members in the former have empty intersection with B , and that in the later have empty intersection with A. $\{I_k\} = \{I'_k\} \sqcup \{I''_k\}$. Thus

$$A \subseteq \bigcup_{k=1}^{\infty} \{I'_k\}, \quad B \subseteq \bigcup_{k=1}^{\infty} \{I''_k\}$$

We have

$$\begin{aligned}\lambda^*(A) + \lambda^*(B) &\leq \sum_{k=1}^{\infty} v(I'_k) + \sum_{k=1}^{\infty} v(I''_k) \\ &= \sum_{k=1}^{\infty} v(I_k) \\ &< \lambda^*(A \cup B) + \varepsilon\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lambda^*(A) + \lambda^*(B) \leq \lambda^*(A \cup B)$$

$\square$

Corollary 1.3.6

All Borel sets in $\mathbb{R}^n$ are Lebesgue measurable. In particular, each open set and each closed set are Lebesgue measurable.

Definition 1.3.7 ▶ Lebesgue Measure

Donote by λ the set function obtained by restricting λ^* to the family of Lebesgue measurable sets. λ is called ***Lebesgue measure***.

Corollary 1.3.8

Let J be an opoen interval in $\mathbb{R}^n$. Then

$$\lambda(J) = \mathrm{vol}(J)$$

Proof. There is a sequence of closed intervals I_k such taht $J = \bigcup_{k=1}^{\infty} I_k$, then

$$\lambda(J) = \lim_{k\to\infty} \lambda(I_k) = \lim_{k\to\infty} \operatorname{vol}(I_k) = \operatorname{vol}(J)$$

□

Theorem 1.3.9

The following five conditions are equivalent for Lebesgue outer measure, λ^*, on $\mathbb{R}^n$:

1. $E \subseteq \mathbb{R}^n$ is λ^*-measurable.
2. For each $\varepsilon > 0$, there is an open set $U \supseteq E$ such that $\lambda^*(U \setminus E) < \varepsilon$.
3. There is a G_δ set $U \supseteq E$ such that $\lambda^*(U \setminus E) = 0$.
4. For each $\varepsilon > 0$, there is a closed set $F \subseteq E$ such that $\lambda^*(E \setminus F) < \varepsilon$.
5. There is an F_σ set $F \subseteq E$ such that $\lambda^*(E \setminus F) = 0$.

Proof. 1. $\Longrightarrow$ 2. : We first assmu that $\lambda^*(E) < \infty$. Take closed intervals $I_1, I_2, \cdots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \lambda^*(E) + \frac{\varepsilon}{2}$$

Let I'_k be closed intervals whose interior contains I_k ,and

$$v(I'_k) < v(I_k) + \frac{\varepsilon}{2^{k+1}}$$

Let

$$U := \bigcup_{k=1}^{\infty} (I'_k)^\circ$$

is a open set.Since U, E is λ-measurable,

$$\begin{aligned}
\lambda^*(U \setminus E) = \lambda(U \setminus E) &= \lambda(U) - \lambda(E) \\
&= \lambda(U) - \lambda^*(E) \\
&< \sum_{k=1}^{\infty} v(I'_k) - \left(\sum_{k=1}^{\infty} v(I_k) - \frac{\varepsilon}{2}\right) \\
&\leq \varepsilon
\end{aligned}$$

Now, note that $\mathbb{R}^n$ is σ-finite, we suppose that $A_1, A_2, \cdots$ are measruable sets

such that
$$\mathbb{R}^n = \bigcup_{j=1}^{\infty} A_j, \quad \lambda\left(A_j\right) < \infty, \forall j \in \mathbb{Z}_{>0}$$

Then $E_j := E \cap A_j$ is a λ^*-measurable set with $\lambda\left(E_j\right) < \infty$ for each j. For each $\varepsilon > 0$, and each $j \in \mathbb{Z}_{>0}$, there is an open set $U_j \supseteq E$ such that
$$\lambda^*\left(U_j \setminus E_j\right) < \frac{\varepsilon}{2^j}$$

Let $U := \bigcup_{j=1}^{\infty} U_j$,then
$$\lambda^*\left(U \setminus E\right) \leq \sum_{j=1}^{\infty} \lambda^*\left(U_j \setminus E\right) \leq \sum_{j=1}^{\infty} \lambda^*\left(U_j \setminus E_j\right) \leq \varepsilon$$

which completes the proof.

2. $\Longrightarrow$ 3. : For each $n \in \mathbb{Z}_{>0}$, there is a open set U_n, such that
$$\lambda^*\left(U_n \setminus E\right) < \frac{1}{n}$$

Let
$$U := \bigcap_{n=1}^{\infty} U_n$$

then U is a G_δ set, such that
$$\lambda^*\left(U \setminus E\right) \leq \lambda^*\left(U_n \setminus E\right) < \frac{1}{n}, \quad \forall n.$$

That is $\lambda^*\left(U \setminus E\right) = 0$.

3. $\Longrightarrow$ 1. : Since λ^* is a Carathéorody measure, then all Borel sets is measurable. Specially, the G_δ set U is measurable. And since λ^*-zero set is measurable, then
$$E = U \setminus \left(U \setminus E\right)$$

is measurable.

2. $\Longrightarrow$ 4. : There is an open set $U \supseteq E^c$ such that $\lambda^*\left(U \setminus E^c\right) < \varepsilon$. Let $F = U^c$, then
$$E \setminus F = E \cap F^c = E \cap U = U \setminus E^c$$

Thus
$$\lambda^*(E \setminus F) = \lambda^*(U \setminus E^c) < \varepsilon$$

4. $\implies$ 5. : For each $n \in \mathbb{Z}_{>0}$, there is a closed set $F_n \subseteq E$ such taht
$$\lambda^*(E \setminus F_n) < \frac{1}{n}$$

Let
$$F := \bigcup_{n=1}^{\infty} F_n,$$

then F is a F_σ set, such that
$$\lambda^*(E \setminus F) \leq \lambda^*(E \setminus F_n) < \frac{1}{n}, \quad \forall n.$$

That is $\lambda^*(E \setminus F) = 0$.

5. $\implies$ 1. : Since $E \setminus F$ and F is measurable, we have
$$E = (E \setminus F) \cup F$$

is measurable. □

Exercises for Lebesgue Measure

Exercise 1.3.10

With λ_t defined by
$$\lambda_t(E) = \lambda(|t|E),$$
prove that $\lambda_t(N) = 0$ whenever $\lambda(N) = 0$.

Proof. Since $\lambda(N) = 0$, for each ε, there is a sequence of closed intervals $I_1, \cdots$ such that
$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \frac{\varepsilon}{|t|^n}$$

where n is the dimension of the Euclidean space we discuss. Then

$$|t|\,E \subseteq \bigcup_{k=1}^{\infty} |t|\,I_k, \quad \sum_{k=1}^{\infty} v\left(|t|\,I_k\right) = |t|^n \sum_{k=1}^{\infty} v\left(I_k\right) < \varepsilon$$

That is

$$\lambda^*\left(|t|\,E\right) < \varepsilon, \quad \forall \varepsilon > 0$$

Thus $|t|\,E$ is measurable, and

$$\lambda\left(|t|\,E\right) = \lambda^*\left(|t|\,E\right) = 0$$

□

Exercise 1.3.11

Let $I, I_1, I_2, \dots, I_k$ be intervals in $\mathbb{R}$ such that $I \subset \bigcup_{i=1}^{k} I_i$. Prove that

$$v(I) \le \sum_{i=1}^{k} v(I_i),$$

where $v(I)$ denotes the length of the interval I.

Proof. We may assume that $v(I) < \infty$, that is I is bounded. Since $v(I) = v\left(\bar{I}\right)$, $v(I_i) = v\left(\bar{I}_i\right)$, and $\bar{I} \subseteq \bigcup_{i=1}^{k} \bar{I}_i$, we may assume that $I, I_1, \cdots, I_k$ are closed intervals. For any $\varepsilon > 0$, there are closed intervals I_i' such that

$$I_i \subseteq \left(I_i'\right)^\circ, \quad v\left(I_i'\right) \le v\left(I_i\right) + \frac{\varepsilon}{2^i}$$

Then $\left\{\left(I_i'\right)^\circ\right\}$ is a open cover of I. Since I is compact, it has a Lebesgue number η. We can subdivide I into subintervals $K_1, \cdots, K_m$ with diameter less that η. Then each K_j is contained in some $(I_{i_j}')^\circ$.

We can formalize this by defining a map $f : \{1, \dots, m\} \to \{1, \dots, k\}$ such that for each j, $K_j \subset (I_{f(j)}')^\circ$. Since the subintervals K_j have disjoint interiors and their union is I, we can write the total length of I and group the sum according to the covering interval index:

$$v(I) = \sum_{j=1}^{m} v(K_j) = \sum_{i=1}^{k} \left(\sum_{j\,:\,f(j)=i} v(K_j) \right).$$

For any fixed i, the intervals $\{K_j\}_{j\,:\,f(j)=i}$ are a collection of disjoint intervals all contained within $(I_i')^\circ$. Therefore, the sum of their lengths cannot exceed the length of $(I_i')^\circ$:

$$\sum_{j\,:\,f(j)=i} v(K_j) = v\left(\bigcup_{j\,:\,f(j)=i} K_j \right) \leq v((I_i')^\circ) = v(I_i').$$

Applying this inequality to our grouped sum, we obtain a single chain of inequalities:

$$v(I) = \sum_{i=1}^{k} \left(\sum_{j\,:\,f(j)=i} v(K_j) \right) \leq \sum_{i=1}^{k} v(I_i') \leq \sum_{i=1}^{k} \left(v(I_i) + \frac{\epsilon}{2^i} \right) = \sum_{i=1}^{k} v(I_i) + \epsilon \sum_{i=1}^{k} \frac{1}{2^i}.$$

Since $\sum_{i=1}^{k} \frac{1}{2^i} < 1$, we have $v(I) < \sum_{i=1}^{k} v(I_i) + \epsilon$. As $\epsilon > 0$ was arbitrary, we conclude that

$$v(I) \leq \sum_{i=1}^{k} v(I_i).$$

□

Exercise 1.3.12

Let $E \subset \mathbb{R}$, and for each real number t, let $E + t = \{x + t \,:\, x \in E\}$. Prove that $\lambda^*(E) = \lambda^*(E + t)$. From this show that if E is Lebesgue measurable, then so is $E + t$.

Proof. Once we show that

$$\lambda^*(E + t) \leq \lambda^*(E), \quad \forall t$$

. By replacing t by $-t$, E by $E + t$, we have

$$\lambda^*(E) \leq \lambda^*(E + t), \forall t$$

Then we get $\lambda^*(E) = \lambda^*(E+t)$. To show this, for each $\varepsilon > 0$, we take closed intervals $I_1, \cdots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \lambda(E) + \varepsilon$$

Then

$$E + t \subseteq \bigcup_{k=1}^{\infty} (I_k + t), \quad \sum_{k=1}^{\infty} v(I_k + t) = \sum_{k=1}^{\infty} v(I_k) < \lambda(E) + \varepsilon.$$

Thus

$$\lambda(E+t) \leq \lambda(E) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, $\lambda(E+t) \leq \lambda(E)$.

To show that $E + t$ is Lebesgue measurable. There is a G_δ set G, such that

$$G \supseteq E, \quad \lambda^*(G \setminus E) = 0.$$

Since open sets is closed under translation and since intersection is kept, then $G_\delta + t$ is a G_δ set as well. We have

$$(G+t) \supseteq (E+t), \quad (G+t) \setminus (E+t) = (G \setminus E) + t$$

Thus

$$\lambda^*((G+t) \setminus (E+t)) = \lambda^*((G \setminus E) + t) = \lambda^*(G \setminus E) = 0$$

Then it follows from Theorem 1.3.9 that $E + t$ is measurable. □

Exercise 1.3.13 ▶ *

Prove that Lebesgue measure on $\mathbb{R}^n$ is independent of the choice of coordinate system. That is, prove that Lebesgue outer measure is invariant under rigid motions in $\mathbb{R}^n$.

Remark.
或许需要积分的替换公式，之后再来看.

Proof. Since we have shown that Lebesgue otuer measure is invariant under translation, we only need to show that is it also invariant under Orthogonal Transformation. □

Exercise 1.3.14

Let P denote an arbitrary $(n-1)$-dimensional hyperplane in $\mathbb{R}^n$. Prove that $\lambda(P) = 0$.

Proof. Since Lebesgue measure is invariant under rigid motions. We may assume that $P = \{x_n = 0\}$. For each $\varepsilon > 0$, let

$$I_k := \left[-\frac{k}{2}, \frac{k}{2}\right]^{n-1} \times \left[\frac{-\varepsilon}{2k^{n-1}} \cdot \frac{1}{2^k}, \frac{\varepsilon}{2k^{n-1}} \cdot \frac{1}{2^k}\right]$$

Then

$$P \subseteq \bigcup_{k=1}^{\infty} I_k, \quad v\left(I_k\right) = \frac{\varepsilon}{2^k}, \quad \forall k$$

Thus

$$\lambda\left(P\right) \le \sum_{k=1}^{\infty} v\left(I_k\right) \le \varepsilon$$

Since ε is arbitrary, $\lambda\left(P\right) = 0$. □

Exercise 1.3.15

In this problem, we want to show that every Lebesgue measurable subset of $\mathbb{R}$ must be "densely populated" in some interval. Thus, let $E \subset \mathbb{R}$ be a Lebesgue measurable set, $\lambda(E) > 0$. For each $\varepsilon > 0$, show that there exists an interval I such that

$$\frac{\lambda(E \cap I)}{\lambda(I)} > 1 - \varepsilon.$$

Note.
我们发现构造这样一个 I 是及其困难的. 转而使用反证法, 去论证 "如果一个建筑的每一块砖都不合格, 那么整个建筑的质量必然有问题".

Proof. Suppose conversely, there exists an $\varepsilon > 0$, such that for each interval I, we have

$$\frac{\lambda\left(E \cap I\right)}{\lambda\left(I\right)} \le 1 - \varepsilon.$$

Let U be a open set such that

$$U \supseteq E, \quad \lambda(U \setminus E) < \delta.$$

Suppose

$$U = \bigcup_{k=1}^{\infty} I_k$$

where I_k are disjoint open intervals. Then

$$\lambda(E \cap U) = \sum_{k=1}^{\infty} \lambda(E \cap I_k) \leq (1-\varepsilon) \sum_{k=1}^{\infty} \lambda(I_k) = (1-\varepsilon)\lambda(U)$$

Since

$$\lambda(U) = \lambda(E \cap U) + \lambda(U \setminus E)$$

, we have

$$\lambda(E \cap U) + \lambda(U \setminus E) \geq \frac{1}{1-\varepsilon}\lambda(E \cap U)$$

Thus

$$\delta \geq \left(1 - \frac{1}{1-\varepsilon}\right)\lambda(E \cap U) = \frac{\varepsilon}{1-\varepsilon}\lambda(E)$$

since $E \cap U = E$. Note that δ can be choosed arbitrarily and indepent with ε and $\lambda(E)$, then it leads to a contradiction, which completes the proof. $\square$

Exercise 1.3.16

Prove that every set $A \subset \mathbb{R}^n$ is contained within a G_δ-set G with the property $\lambda(G) = \lambda^*(A)$.

Proof. For any $n > 0$, there are closed intervals $I_1^{(n)}, I_2^{(n)}, \cdots$ such that

$$A \subseteq \bigcup_{k=1}^{\infty} I_k^{(n)}, \quad \sum_{k=1}^{\infty} v\left(I_k^{(n)}\right) \leq \lambda^*(A) + \frac{1}{2n}$$

For each k, there is a closed interval $\left(I_k^{(n)}\right)'$ such that

$$I_k^{(n)} \subseteq \left(\left(I_k^{(n)}\right)'\right)^{\circ}, \quad v\left(\left(I_k^{(n)}\right)'\right) < v\left(I_k^{(n)}\right) + \frac{1}{2^{k+1}n}$$

Let

$$G := \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{\infty} \left(\left(I_k^{(n)} \right)' \right)^{\circ}$$

is a G_δ set. Then

$$\lambda(G) \le \lambda\left(\bigcup_{k=1}^{\infty} \left(I_k^{(n)} \right)' \right) \le \sum_{k=1}^{\infty} v\left(\left(I_k^{(n)} \right)' \right) < \lambda^*(A) + \frac{1}{n}, \quad \forall n$$

Thus

$$\lambda(G) \le \lambda^*(A)$$

The other hand, since $G \supseteq A$, it is immediate that

$$\lambda(G) = \lambda^*(G) \ge \lambda^*(A).$$

Finally,

$$\lambda(G) = \lambda^*(A)$$

□

Exercise 1.3.17

Let $\{E_k\}$ be a sequence of Lebesgue measurable sets contained in a compact set $K \subset \mathbb{R}^n$. Assume for some $\varepsilon > 0$ that $\lambda(E_k) > \varepsilon$ for all k. Prove that there is some point that belongs to infinitely many E_k's.

Note.
对于由一列成员组成的族类，一个点属于无穷多个成员，当且仅当它永远无法仅被有限多个成员“杀死”，当且仅当它是“反复出现”的，当且仅当它属于集合列的极限上集.

Proof. A point belongs to infinitely many E_k, if and only if it belongs to

$$\limsup_{k\to\infty} E_k := \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$$

Since $\lambda\left(\bigcup_{k=1}^{\infty} E_k\right) \le \lambda(K) < \infty$. It follows from Corollary 1.1.11 that

$$\lambda\left(\limsup_{k\to\infty} E_k \right) \ge \limsup_{k\to\infty} \lambda(E_k) \ge \varepsilon > 0$$

Thus

$$\lim_{k\to\infty} \sup E_k \neq \varnothing$$

which completes the proof. □

Exercise 1.3.18

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a Lipschitz map. Prove that if $\lambda(E) = 0$, then $\lambda(T(E)) = 0$.

Proof. Suppose that

$$d(T(x_1), T(x_2)) \leq L \cdot d(x_1, x_2)$$

Since $\lambda(E) = 0$, for any $\varepsilon > 0$, there are closed intervals $I_1, \cdots$, such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \varepsilon$$

There are finitely many Cubes $Q_{k1}, \cdots, Q_{km_k}$, such that

$$I_k \subseteq \bigcup_{i=1}^{m_k} Q_{ki}, \quad \sum_{i=1}^{m_k} v(Q_{ki}) < 2v(I_k)$$

Thus we may assume that I_k are all cubes. Denote the diameter of I_k by $\sqrt{n}a_k$, and denote the centre of I_k by c_k, then

$$d(T(x), T(c_k)) \leq L \cdot d(x, c_k) \leq \sqrt{n}L \cdot a_k$$

Thus $T(I_k)$ is contained by the cube centering $T(c_k)$ with diameter $nL \cdot a_k$, which we denote by I'_k, then

$$v\left(I'_k\right) \leq n^n L^n v(I_k)$$

. We have

$$T(E) \subseteq \bigcup_{k=1}^{\infty} I'_k, \quad \sum_{k=1}^{\infty} v\left(I'_k\right) < n^n L^n \varepsilon$$

, which concludes that $\lambda(T(E)) = 0$. □

1.4 The Contor Set

Definition 1.4.1 ▶ Contor set

The Cantor set is a subset of the interval $[0,1]$. We will desccribe it by constructing its complement in $[0,1]$.

1. At the fist step, let $I_{1,1}$ denote the open interval $\left(\frac{1}{3},\frac{2}{3}\right)$. Thus $I_{1,1}$ is the open middle third of the interval $I=[0,1]$.
2. The second step involves performing the first tep on each of the two remaining intervals of $I-I_{1,1}$. That is, We produce two open intervals, $I_{2,1}$ and $I_{2,2}$, each being the open middle third of one of the two intervals that constitute $I-I_{1,1}$.
3. At the ith step we produce 2^{i-1} open intervals $I_{i,1},I_{i,2},\cdots,I_{i,2^{i-1}}$, each of length $\left(\frac{1}{3}\right)^i$. The $(i+1)$ th step consists in producing mid-ddle thirds of each of the intervals of

$$I-\bigcup_{j=1}^{i}\bigcup_{k=1}^{2^{j-1}} I_{j,k}.$$

4. With C denoting the Cantor set, we define its complement by

$$I\setminus C=\bigcup_{j=1}^{\infty}\bigcup_{k=1}^{2^{j-1}} I_{j,k}.$$

Proposition 1.4.2

Let C be the Contor set. Then the following holds:

1. C is closed, that is $C=\bar{C}$.
2. Let λ be the Lebesgue measure, then

$$\lambda\left(\bar{C}\right)=\lambda\left(C\right)=0$$

3. C is nowhere dense.
4. The cardinality of C is

$$|C|=2^{\aleph_0}=|R|=c$$

Proof. 1. $I\setminus C$ is open by definition. Thus C is closed.

2.
$$\lambda(I \setminus C) = \sum_{j=1}^{\infty} \sum_{k=1}^{2^{j-1}} \lambda\left(I_{j,k}\right) = \sum_{j=1}^{\infty} 2^{j-1}\left(\frac{1}{3}\right)^{j} = \frac{1}{2}\sum_{j=1}^{\infty}\left(\frac{2}{3}\right)^{j} = 1$$

Thus
$$\lambda(C) = 1 - \lambda(I \setminus C) = 1 - 1 = 0$$

3.
> *Note.*
> [0, 1] 上的每一个值通过修饰存在唯一的二进制表示. 同样地也存在唯一的三进制表示. Contor 集就是把每个三进制表示中, 字符含 1 的数都去掉.

Every number $x \in [0,1]$ has a ternary expansion of the form

$$x = \sum_{i=1}^{\infty} \frac{x_i}{3^i}$$

where each x_i is 0, 1, or 2 and we write $x = 0, x_1 x_2 \cdots$ This expansion unique except when
$$x = \frac{a}{3^n}$$
(Since $\frac{1}{3^n}$ has another expresion $\sum_{i=n+1}^{\infty} \frac{2}{3^i}$)where a and n are positive integers with $0 < a < 3^n$ and where 3 does not divide a. In this case, x has the form
$$x = \sum_{i=1}^{n} \frac{x_i}{3^i}$$

where x_i is either 1 or 2. If $x_n = 2$, we will use this expression to represent x. However, if $x_n = 1$, we will use the following representation for x:
$$x = \frac{x_1}{3} + \frac{x_2}{3^2} + \cdots + \frac{x_{n-1}}{3^{n-1}} + \frac{0}{3^n} + \sum_{i=n+1}^{\infty} \frac{2}{3^i}.$$

Thus , with this convention, each number $x \in [0,1]$ has a uniue ternary expansion.

Let $x \in I$ and conssider its ternary expansion $x = 0.x_1 x_2 \cdots$, bearing in mind the convention we have adopted above. Obseve that $x \notin I_{1,1}$ if and only if $x_1 \neq 1$. Also , if $x_1 \neq 1$, then $x \notin I_{2,1} \cup I_{2,2}$ if and only if $x_2 \neq 1$. Continuing in this way, we see that $x \in C$ if and only if $x_i \neq 1$ for each positive integer i. Thus, there is a one-to-one correspondence between elements of C and all sequences $\{x_i\}$ in which each x_i is either 0 or 2. The cardinality of the latter is $2^{\aleph_0}$, which is $|\mathbb{R}| = c$.

$\square$

Exercises for Cantor set

Exercise 1.4.3

Prove that the Cantor-type set C_α (The method of construction is the same as C,except that at the ith step, each of the intervals removed has length $\alpha 3^{-i}$) is nowhere dense, has cardinality c, and has Lebesgue measure $1-\alpha$.

Proof.

- $\overline{C}_\alpha = C_\alpha$. Only need to show that

$$\operatorname{int}(C_\alpha) = \emptyset$$

With E_k denoting the result of the kth step constructing the C_α, we define its complements by

$$I \setminus E_k := \bigcup_{j=1}^{k} \bigcup_{m=1}^{2^{j-1}} I_{m,j}$$

Then $E_1 \supseteq E_2 \supseteq \cdots$, and $C_\alpha = \bigcap_{k=1}^{\infty} E_k$.If there is a nonempty open inverval $(a,b) \subseteq C_\alpha$. Then

$$(a,b) \subseteq E_k, \quad \forall k$$

Note that E_k is a disjoint union of closed intervals with length l_k, which satisfies

$$l_k = \frac{1}{2}\left(l_{k-1} - \alpha_3^{-k}\right) < \frac{1}{2} l_{k-1} \implies l_k < \frac{1}{2^k}$$

. There is a k_0 such that $l_{k_0} < b-a$. Since

$$(a,b) \subseteq E_{k_0}$$

is a connected set lying in E_{k_0}, E_{k_0} has a component with length more than $b-a$, which is a contradiction. Thus there is no open interval contained in C_α, that is $C_\alpha = \bar{C}_\alpha$ is no where dense.

- We denote the two closed intervals of $I \setminus I_{1,1}$ by J_1, J_2. Denote the two closed intervals of $J_1 - I_{2,1}$ by $J_{1,1}, J_{1,2}$, and denote the two that of $J_2 - I_{2,2}$ by $J_{2,1}, J_{2,2}$. Inductively, denote the two intervals of $J_{i_1,i_2,\cdots,i_{j-1}} - I_{j,2^{j-1}i_1+\cdots+2^0 i_{j-1}}$ by $J_{i_1,\cdots,i_{j-1},1}, J_{i_1,\cdots,i_{j-1},2}$. Then

$$E_k = \bigcup_{i_1,\cdots,i_k \in \{1,2\}^k} J_{i_1,\cdots,i_k}$$

Let

$$\Sigma := \{1,2\}^{\mathbb{N}}$$

Given a $(s_1, s_2, \cdots) \in \Sigma$, we can get a sequence of nested closed intervals

$$J_{s_1} \supseteq J_{s_1,s_2} \supseteq \cdots$$

From Nested Interval Theorem, there is a unique point $s \in \bigcap_{k=1}^{\infty} J_{s_1,\cdots,s_k}$. We define a f by mapping $(s_1, s_2, \cdots) \in \Sigma$ to $s \in C_\alpha$. To show that f is injective, we take

$$s = (s_1, s_2, \cdots), t = (t_1, t_2, \cdots) \in \Sigma, \quad s \neq t$$

Then there is a smallest $k \in \mathbb{Z}_{>0}$, such that $s_k \neq t_k$. We may assume that $s_k = 1$, $t_k = 2$. Then

$$f(s) \in J_{s_1,s_2,\cdots,s_{k-1},1}, \quad f(t) \in J_{t_1,t_2,\cdots,t_{k-1},2} = J_{s_1,s_2,\cdots,s_{k-1},2}$$

Since $J_{s_1,s_2,\cdots,s_{k-1},1} \cap J_{s_1,s_2,\cdots,s_{k-1},2} = \varnothing$, $f(s) \neq f(t)$. Thus f is injective. To show that f is surjective. Take any $x \in C_\alpha$. Since

$$C_\alpha = \bigcap_{k=1}^{\infty} E_k$$

, we have

$$x \in E_k, \quad \forall k$$

And since

$$E_1 = J_1 \cup J_2$$

is a disjoint union. There exists a unique $i_1 \in \{1,2\}$, such that

$$x \in J_{i_1}$$

Then
$$x \in E_2 \cap J_{i_1} = J_{i_1,1} \cup J_{i_1,2}$$
There exists a unique $i_2 \in \{1, 2\}$ such that
$$x \in J_{i_1,i_2}$$
Inductively, once we show that there exists a unique $(i_1, \cdots, i_k) \in \{1, 2\}^k$ such that
$$x \in J_{i_1,i_2,\cdots,i_k}$$
From
$$x \in E_{k+1} \cap J_{i_1,\cdots,i_k} = J_{i_1,\cdots,i_k,1} \cup J_{i_1,\cdots,i_k,2}$$
we know that there exists a unique $(i_1, \cdots, i_{k+1}) \in \{1, 2\}^{k+1}$, such that
$$x \in J_{i_1,\cdots,i_{k+1}}$$
Thus we have determined a unique $(i_1, i_2, \cdots) \in \Sigma$, such that
$$x \in \bigcap_{k=1}^{\infty} J_{i_1,i_2,\cdots,i_k}$$
That is
$$f(i_1, i_2, \cdots) = x$$
Thus f is surjective. Finally, we have
$$c = 2^{\aleph_0} = |\Sigma| = |C_\alpha|$$

- $$\lambda(I \setminus C_\alpha) = \lambda\left(\bigcup_{j=1}^{\infty} \bigcup_{k=1}^{2^{j-1}} I_{k,j}\right) = \sum_{j=1}^{\infty} 2^{j-1}\alpha\left(\frac{1}{3}\right)^j = \alpha$$

Thus
$$\lambda(C_\alpha) = \lambda(I) - \lambda(I \setminus C_\alpha) = 1 - \alpha$$

□

Exercise 1.4.4

Construct an open set $U \subset [0,1]$ such that U is dense in $[0,1]$, $\lambda(U) < 1$, and $\lambda(U \cap (a,b)) > 0$ for every interval $(a,b) \subset [0,1]$.

Proof. Define

$$U := I \setminus C_\alpha, \quad 0 < \alpha < 1$$

Where C_α is the Cantor type set with Lebesgue measure $1-\alpha$. Then

$$\lambda(U) = \lambda(I) - \lambda(C_\alpha) = \alpha > 0$$

If U is not dense. Then there is a nonempty open subset $Y \subseteq [0,1]$ such that

$$Y \subseteq U^c = C_\alpha$$

which is a contradicion to C_α is no where dense. Since U is dense,

$$U \cap (a,b) \neq \varnothing$$

That is $U \cap (a,b)$ is a nonempty open set, thus its Lebesgue measure is nonzero. □

Exercise 1.4.5

Prove that the family of Borel subsets of $\mathbb{R}$ has cardinality c. From this deduce the existence of a Lebesgue measurable set that is not a Borel set.

Exercise 1.4.6

Let E be the set of numbers in $[0,1]$ whose ternary expansions have only finitely many 1's. Prove that $\lambda(E) = 0$.

Proof. Let E_k be the set of numbers in $I = [0,1)$ whose ternary expansions have no $1's$ after the kth digit . Then

$$E = \bigcup_{k=1}^{\infty} E_k, \quad E_1 \subseteq E_2 \subseteq \cdots$$

Let I_1, I_2, I_3 be the fomer, middle, latter third of the $I = [0,1)$ respectively, specially, we say $[0, \frac{1}{3}), [\frac{1}{3}, \frac{2}{3}), [\frac{2}{3}, 1)$. Once we have defined $I_{i_1,i_2,\cdots,i_k}$, then we define $I_{i_1,i_2,\cdots,i_k,1}$ by the first third of the interval $I_{i_1,i_2,\cdots,i_k}$ as before , the remainings are similar. We have

$$E_k = I \setminus \bigcup_{j=k}^{\infty} \bigcup_{i_1,i_2,\cdots,i_k \in \{1,2,3\}^k, i_{k+1},\cdots,i_j \in \{1,3\}^{j-k}} I_{i_1,i_2,\cdots,i_j,2}$$

Let

$$A_{k,m} := \bigcup_{j=k}^{m} \bigcup_{i_1,i_2,\cdots,i_k \in \{1,2,3\}^k, i_{k+1},\cdots,i_j \in \{1,3\}^{j-k}} I_{i_1,i_2,\cdots,i_j,2}$$

Then

$$A_{k,k} \subseteq A_{k,k+1} \subseteq \cdots$$

Denote $\bigcup_{m=k}^{\infty} A_{k,m}$ by A_k, then $E_k = I \setminus A_k$. We have

$$\lambda(A_k) = \lim_{m\to\infty} \lambda(A_{k,m}) = \sum_{j=k}^{\infty} 3^k 2^{j-k} \left(\frac{1}{3}\right)^{j+1} = \frac{1}{3}\left(\frac{3}{2}\right)^k \sum_{j=k}^{\infty} \left(\frac{2}{3}\right)^j = 1$$

Thus

$$\lambda(E_k) = \lambda(I) - \lambda(A_k) = 1 - 1 = 0$$

It follows from the continuity of λ from below that

$$\lambda(E) = \lim_{k\to\infty} \lambda(E_k) = 0$$

□

Proof. Another proof, use Cantor set.

Note.
- 标准 Cantor 集合描述了 [0,1] 内三进制表示不含 1 的那些数.
- 通过对 Cantor 集 $\frac{1}{3^k}$ 倍的缩放, 可以实现所描述的三进制范围表示相对于小数点的滞后 (k 个字符).
- 缩放后 Cantor 集的起点, 决定了新 Cantor 集开始作用的之前的三进制字符都是哪些.
- 一切的一切, 源于 Cantor 集的自相似性, 也就是说

$$C = \left(\frac{1}{3}C\right) + \left(\frac{1}{3}C + \frac{2}{3}\right)$$

E_k is defined as above. Let $P_k := \{0,1,2\}^k$. For $p = (d_1, \cdots, d_k) \in P_k$, define

$$C_p := \frac{1}{3^k}C + \sum_{j=1}^{k} \frac{d_j}{3^j}$$

Then the similar process we describe C gives that

$$C_p = \left\{x \in [0,1] : x = (0.m_1 m_2 \cdots)_{ternary}, m_i = d_i \text{ for } 1 \leq i \leq k, m_j \neq 1 \text{ for } j > k\right\}$$

We have

$$E_k = \bigcup_{p \in P_k} C_p$$

Since

$$\lambda\left(C_p\right) = \frac{1}{3^k}\lambda(C) = 0$$

We have

$$E_k \leq \sum_{p \in P_k} \lambda\left(C_p\right) = 0$$

Then

$$\lambda(E) = \lim_{k \to \infty} \lambda(E_k) = 0$$

□

1.5 Existence of Nonmeasurable Sets

Lemma 1.5.1

If $S \subset \mathbb{R}$ is a Lebesgue measurable set of positive and finite measure, then the set of differences $D_S := \{x - y : x, y \in S\}$ contains an interval.

Proof sketch.
Since

$$\begin{aligned} x \in D_S &\iff S \cap (S + x) \neq \varnothing \\ &\iff \text{There exists } T \subseteq S \text{ such that } T \cap (T + x) \neq \varnothing \end{aligned}$$

Only need to find a such T, and a interval I, such that $T \cap (T + x) \neq \varnothing$ for $\forall x \in I$.

> When $T \cap (T+x) = \varnothing$, we have $\lambda(T \cup (T+x)) = 2\lambda(T)$. Once we find a not too large (less than $2\lambda(T)$) set containing $T \cup (T+x)$ for all $x \in I$. From another perspective, we need to find an T occupise at least half of a set (relative to Lebesgue measure) and dose not escape it under a tiny transformation.

Proof. For each $\varepsilon > 0$, there exists an open set $U \supseteq S$ with $\lambda(U) < (1+\varepsilon)\lambda(S)$. We write U as a countable disjoint union of open intervals:

$$U = \bigcup_{k=1}^{\infty} I_k$$

Then

$$\lambda(S) = \lambda(S \cap U) = \sum_{k=1}^{\infty} \lambda(S \cap I_k)$$

We have

$$\sum_{k=1}^{\infty} \lambda(I_k) < (1+\varepsilon) \sum_{k=1}^{\infty} \lambda(S \cap I_k)$$

Then there is some k_0 such that

$$\lambda(I_k) < (1+\varepsilon)\lambda\left(S \cap I_{k_0}\right)$$

We take $\varepsilon = \frac{1}{3}$, then

$$\lambda\left(S \cap I_{k_0}\right) > \frac{3}{4}\lambda\left(I_{k_0}\right)$$

For any $|t| < \frac{1}{2}\lambda\left(I_{k_0}\right)$, $S \cap I_{k_0} \cup \left(S \cap I_{k_0} + t\right)$ is contained in some open interval with Lebesgue measure $\frac{3}{2}\lambda\left(I_{k_0}\right)$. Thus

$$\lambda\left(S \cap I_{k_0} \cup \left(S \cap I_{k_0} + t\right)\right) \leq \frac{3}{2}\lambda\left(I_{k_0}\right)$$

Then it follows that it is impossible that

$$\left(S \cap I_{k_0}\right) \cap \left(S \cap I_{k_0} + t\right) = \varnothing, \quad \text{for some } |t| < \frac{1}{2}$$

since otherwise the Lebsgue measure of the union of the two is greater than

$2\lambda\left(S \cap I_{k_0}\right) > \frac{3}{2}\lambda\left(I_{k_0}\right)$, which is a contradiction. Now we have

$$\left(-\frac{1}{2}, \frac{1}{2}\right) \subseteq D_{S \cap I_{k_0}} \subseteq D_S$$

□

Exercise 1.5.2

There exists a set $E \subseteq \mathbb{R}$ that is not Lebesgue measurable.

Proof. We define a ralation on the real line by

$$x \sim y :\iff x - y \in \mathbb{Q}$$

Then it is easy to verify that $\sim$ is a equivalent relation. Then $\mathbb{R}$ can be seprated by the class of the relation. Note that every equivalent class is countable. But $\mathbb{R}$ is uncountable, there are uncountably many equivalent class. Now we use the axiom of choice, to get a set S by selecting precisely one point from each equivalent class. Then

$$\mathbb{R} = S + \mathbb{Q} := \bigcup_{r \in \mathbb{Q}} (S + r)$$

If S is Lebesgue measurable, then $\lambda(S) > 0$ since

$$\infty = \lambda(\mathbb{R}) = \sum_{r \in \mathbb{Q}} \lambda(S + r) = \sum_{r \in \mathbb{Q}} \lambda(S)$$

Now, considering

$$D_S = \{x - y : x, y \in S\}$$

If $x \neq y$, since x, y belong to different class, $x - y \notin \mathbb{Q}$. Then we have

$$D_S \subseteq 0 \cup (\mathbb{R} \setminus \mathbb{Q})$$

which cannot contain any inverval. Then it follows from the lemma above that $\lambda(S) = \infty$. But then there exists a closed inverval I such that

$$0 < \lambda(S \cap I) < \infty$$

and $S \cap I$ is measurable, it follows fome the lemma again that there is a interval

$$J \subseteq D_{S \cap I} \subseteq D_S$$

, which is a contradiction. □

Exersices for Existence of Nonmeasurable Sets

Exercise 1.5.3

Prove that there every subset of $\mathbb{R}$ with positive outer Lebesgue measure contains a nonmeasurable subset.

Proof. Let $T \subseteq \mathbb{R}$ be any subset of $\mathbb{R}$ with positive outer Lebesgue measure. The $\sim$ defined as above is also a equivalent relation on T. Let S be the same set we just defined in the above theorem.

$$T = T \cap \mathbb{R} = T \cap \left(\bigcup_{r \in \mathbb{Q}} (S + r) \right) = \bigcup_{r \in \mathbb{Q}} (T \cap (S + r))$$

which is a disjoint union. We have

$$0 < \lambda^* (T) \leq \sum_{r \in \mathbb{Q}} \lambda^* (T \cap (S + r))$$

There exists an $r_0 \in \mathbb{Q}$ such that

$$\lambda^* (T \cap (S + r_0)) > 0$$

We denote $T \cap (S + r_0)$ by S'. If S' is Lebesgue measurable. Note that $D_{S'} \subseteq (\mathbb{R} \setminus \mathbb{Q}) \cup \{0\}$. If $\lambda (S') < \infty$, then it is a contradiction to Lemma 1.5.1. If $\lambda (S') = \infty$. Then there exists a closed interval I such that

$$0 < \lambda (S' \cap I) < \infty$$

then there is an interval J such that

$$J \subseteq D_{S' \cap I} \subseteq D_{S'}$$

which is a contradicion as well. □

Exercise 1.5.4

Let $\mathcal{N}$ denote the nonmeasurable set constructed in this section. Show that if E is a measurable subset of $\mathcal{N}$, then $\lambda(E) = 0$.

Proof. We still use the notation S to represent that nonmeasurable set. If

$$E \subseteq S, \quad \lambda(E) > 0$$

If $\lambda(E) < \infty$, then from Lemma 1.5.1 there is an interval J such that

$$J \subseteq D_E \subseteq D_S$$

which is a contradiction. Again, if $\lambda(E) = \infty$, then there exists a closed interval I such that

$$0 < \lambda(E \cap I) < \infty$$

Then again from Lemma 1.5.1 there exists an interval J such that

$$J \subseteq D_{E\cap I} \subseteq D_E \subseteq D_S,$$

which is a contradicion. □

1.6 Lebesgue-Stieljes Measure

Definition 1.6.1

If f is a nondecreasing function defined on $\mathbb{R}$, then the "length" of a half-open interval $(a,b]$, denoted by $\alpha_f((a,b])$, can be defined by

$$\alpha_f((a,b]) = f(b) - f(a)$$

Definition 1.6.2 ▸ Lebesgue-Stieltjes outer measure

The Lebesgue-Stieltjes outer measure of an arbitrary set $E \subseteq \mathbb{R}$ is defined by

$$\lambda_f^*(E) := \inf\left\{\sum_{h_k\in\mathcal{F}} \alpha_f(h_k)\right\}$$

where the infimum is take over all countable collections $\mathcal{F}$ of half-open intervals h_k of the form $(a_k, b_k]$ such that

$$E \subseteq \bigcup_{h_k\in\mathcal{F}} h_k$$

Remark.
If f is right-continuous, then the (Lebesgue) length of each interval $(a_k, b_k]$ that appears above can be assumed to be arbitrarily small, beacause

$$\alpha_f((a,b]) = f(b) - f(a) = \sum_{k=1}^{N} [f(a_k) - f(a_{k-1})] = \sum_{k=1}^{N} \alpha_f((a_{k-1}, a_k])$$

whenever $a = a_0 < a_1 < \cdots < a_N = b$.

Theorem 1.6.3

If $f : \mathbb{R} \to \mathbb{R}$ is a nondecreasing function, then λ_f^* is a Carathéodory outer measure. We denote the restriction of λ_f^* on measruable sets by λ_f.

Proof. To show that λ_f^* is a outer measure, we only need to show the it is monotone and countably subadditive.

For the proof of monotonicity, let $A_1 \subseteq A_2$ be any subset of $\mathbb{R}$. Take any $\varepsilon > 0$. There is a countable collections $h_1, h_2, \cdots$ of $\mathcal{F}$, such that

$$A_2 \subseteq \bigcup_{k=1}^{\infty} h_k, \quad \sum_{k=1}^{\infty} \alpha_f(h_k) < \lambda_f^*(A_2) + \varepsilon$$

Then

$$A_1 \subseteq \bigcup_{k=1}^{\infty} h_k, \quad \sum_{k=1}^{\infty} \alpha_f(h_k) < \lambda_f^*(A_2) + \varepsilon$$

That is

$$\lambda_f^*(A_1) \leq \lambda_f^*(A_2) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, $\lambda_f^*(A_1) \leq \lambda_f^*(A_2)$.

The proof of countable subadditivity is virtually identical to the proof the corresponding result of Lebesgue measure given in Theorem 1.3.5, since we didn't use any explicit expresion of the length of an interval there.

The same idea of prooving Lebesgue outer measure to be Carathéodory is effective for that of the Lebesgue-Stieljes oute measure, since the only thing we should check to repeate that proof here is whether we can subdividing each h_k into smaller members of $\mathcal{F}$ whose diameter less than $\mathbf{d}(A,B)$. But that was shown in the remark of the definition. □

Theorem 1.6.4

If $f:\mathbb{R}\to\mathbb{R}$ is nondecreasing and right-continuous then

$$\lambda_f((a,b]) = f(b) - f(a)$$

Note.
回忆 Lebesgue 测度的情况，我们将闭集构成的覆盖，通过微小的几乎不影响测度的 (向两边) 扩张，其内部形成一组开覆盖. 我们是利用这组开覆盖来覆盖身为紧集的闭区间，以处理问题的.
这里，我们由于选择了半开区间，只需要对右侧进行微小的扩张，作为代价，我们研究的集合本身不再是紧集. 这里假设 f 是右连续，就同时解决了两个问题:

1. 向右微小扩张，得益于右连续，对测度的影响也变成是无限小的.
2. 得益于右连续，用闭集 $[a',b]$，向左逼近 $(a,b]$, $a'\to a^+$ 的行为也是良好的.

Proof. It suffices to show that

$$\lambda_f((a,b]) \geq f(b) - f(a)$$

For each $\varepsilon>0$, we take a sequence of half-invervals $h_1=(a_1,b_1], h_2=(a_2,b_2],\cdots$ such that

$$(a,b] \subseteq \bigcup_{k=1}^{\infty} h_k, \quad \sum_{k=1}^{\infty} \alpha_f(h_k) < \lambda_f((a,b]) + \varepsilon.$$

Since f is right-continuous , we have

$$\lim_{t\to 0^+} \alpha_f((a_k, b_k+t]) = \alpha_f((a_k,b_k])$$

Consequently, we may replace each $(a_k,b_k]$ with $(a_k,b'_k]$ such that

$$b_k < b'_k, \quad \alpha_f((a_k,b'_k]) < \alpha_f((a_k,b_k]) + \frac{\varepsilon}{2^k}$$

Let $a'\in(a,b)$. The compact set $[a',b]$ is covered by a finite family of open intervals, which we denote $\mathcal{C}_{fin}=\{C_j\}_{j=1}^N$, where $C_j=(a_j,b'_j)$. $[a',b]\subseteq$

$\bigcup_{j=1}^{N} C_j$. By the Lebesgue number lemma, there exists an $\eta > 0$ such that any subset of $[a', b]$ with diameter less than η is contained in at least one C_j. We partition $[a', b]$ into subintervals $I_k = [t_{k-1}, t_k]$ for $k = 1, \ldots, m$, each with length less than η. Thus, for each I_k, we can choose an interval from $\mathcal{C}_{fin}$ that contains it.

Let's formalize this choice. We define an assignment function $\phi : \{1, \ldots, m\} \to \{1, \ldots, N\}$ such that for each k, $I_k \subseteq C_{\phi(k)}$.

We can now write the total increase of f over $[a', b]$ and group the terms according to our assignment:

$$f(b) - f(a') = \sum_{k=1}^{m} (f(t_k) - f(t_{k-1})) = \sum_{j=1}^{N} \left(\sum_{k \text{ s.t. } \phi(k)=j} (f(t_k) - f(t_{k-1})) \right)$$

The outer sum is over the indices of the open cover $\mathcal{C}_{fin}$, ensuring no open interval is counted more than once.

Now, let's analyze the inner sum for a fixed j. Let $\mathcal{P}_j = \{I_k \mid \phi(k) = j\}$ be the set of subintervals assigned to C_j. These are a finite number of disjoint subintervals, all contained within $C_j = (a_j, b'_j)$. Let $t_{\min,j}$ be the minimum of all left endpoints of intervals in $\mathcal{P}_j$, and $t_{\max,j}$ be the maximum of all right endpoints. Then $\bigcup_{I_k \in \mathcal{P}_j} I_k \subseteq [t_{\min,j}, t_{\max,j}]$. Since f is nondecreasing, the sum of increases over disjoint intervals is less than or equal to the increase over their convex hull:

$$\sum_{I_k \in \mathcal{P}_j} (f(t_k) - f(t_{k-1})) \leq f(t_{\max,j}) - f(t_{\min,j})$$

Furthermore, since all these I_k are in $C_j = (a_j, b'_j)$, we have $a_j < t_{\min,j}$ and $t_{\max,j} < b'_j$. The nondecreasing property of f implies:

$$f(t_{\max,j}) - f(t_{\min,j}) \leq f(b'_j) - f(a_j)$$

Combining these, for each j, we have established:

$$\sum_{k \text{ s.t. } \phi(k)=j} (f(t_k) - f(t_{k-1})) \leq f(b'_j) - f(a_j)$$

Substituting this back into our grouped sum:

$$f(b) - f(a') \leq \sum_{j=1}^{N} (f(b'_j) - f(a_j))$$

This sum is over a finite subset of the original infinite cover used to define the 2ε bound. Therefore:

$$\sum_{j=1}^{N} (f(b'_j) - f(a_j)) \leq \sum_{i=1}^{\infty} (f(b'_i) - f(a_i)) < \lambda_f((a, b]) + 2\varepsilon$$

So we have $f(b) - f(a') < \lambda_f((a, b]) + 2\varepsilon$.

Since ε is arbitrary, we have $f(b) - f(a') \leq \lambda_f((a, b])$. Furthermore, the right continuity of f implies $\lim_{a' \to a^+} f(a') = f(a)$, and hence

$$f(b) - f(a) \leq \lambda_f((a, b])$$

as desired. □

Theorem 1.6.5

Suppose μ is a finite Borel outer measure on $\mathbb{R}$ and let

$$f(x) = \mu((-\infty, x]).$$

Then the Lebesgue-Stieltjes measure, λ_f, agrees with μ on all Borel sets.

Proof. * f is nondecreasing from definition. Note that

$$(-\infty, x] = \bigcap_{n \in \mathbb{N}} (-\infty, x + \frac{1}{n}]$$

Since μ is finite, the continuity from below gives that

$$\lim_{n \to \infty} \mu\left((-\infty, x + \frac{1}{n}]\right) = \mu\left((-\infty, x]\right)$$

That is

$$\lim_{n \to \infty} f\left(x + \frac{1}{n}\right) = f\left(x\right)$$

Since f is nondecreasing ,

$$\lim_{t\to 0^+} f(x+t) = f(x)$$

Since

$$(-\infty, a] \cup (a, b] = (-\infty, b]$$

then

$$f(b) - f(a) = \mu((-\infty, b]) - \mu((-\infty, a]) = \mu((a, b])$$

thus proving that μ and λ_f agree on all half-open intervals. Since very open set in $\mathbb{R}$ is a countable union of disjoint half-open intervals, it follows that μ and λ_f agree on all open sets. It will be shown later that μ, λ_f agree on all Borel sets . □

Exercises for Lebesgue-Stieltjes Measure

Exercise 1.6.6

Let $f : \mathbb{R} \to \mathbb{R}$ be a nondecreasing function and let λ_f be the Lebesgue-Stieltjes measure generated by f. Prove that $\lambda_f(\{x_0\}) = 0$ if and only if f is left-continuous at x_0.

Proof.

$$\{x_0\} = \bigcap_{n\in\mathbb{N}} (x_0, x_0 + \frac{1}{n}]$$

It follows from the continuity from above that

$$\lim_{n\to\infty} \left(f\left(x_0 + \frac{1}{n}\right) - f(x_0) \right) = \lambda_f(\{x_0\})$$

Since f is nondecreasing,

$$\lim_{t\to 0^+} f(x_0+t) = \lim_{n\to\infty} f\left(x_0 + \frac{1}{n}\right)$$

We have

$$\lim_{t\to 0^+} f(x_0+t) - f(x_0) = \lambda_f(\{x_0\})$$

The LHS= 0 iff f is left-continuous, which completes the proof. □

Exercise 1.6.7

Let f be a nondecreasing function defined on $\mathbb{R}$. Define a Lebesgue-Stieltjes-type measure as follows: For $A \subset \mathbb{R}$ an arbitrary set,

$$\Lambda_f^*(A) = \inf\left\{\sum_{h_k \in \mathcal{F}} [f(b_k) - f(a_k)]\right\},$$

where the infimum is taken over all countable collections $\mathcal{F}$ of closed intervals of the form $h_k := [a_k, b_k]$ such that

$$E \subset \bigcup_{h_k \in \mathcal{F}} h_k.$$

In other words, the definition of $\Lambda_f^*(A)$ is the same as $\lambda_f^*(A)$ except that closed intervals $[a_k, b_k]$ are used instead of half-open intervals $(a_k, b_k]$. As in the case of Lebesgue-Stieltjes measure it can be easily seen that Λ_f^* is a Carathéodory measure. (You need not prove this.)

(a) Prove that $\Lambda_f^*(A) \leq \lambda_f^*(A)$ for all sets $A \subset \mathbb{R}^n$.

(b) Prove that $\Lambda_f^*(B) = \lambda_f^*(B)$ for all Borel sets B if f is left-continuous.

Proof. 1. Let $(a_1, b_1], (a_2, b_2], \cdots$ be a sequence of half-open intervals , such that

$$A \subseteq \bigcup_{k=1}^{\infty}(a_k, b_k], \quad \sum_{k=1}^{\infty}(f(b_k) - f(a_k)) < \lambda_f^*(A) + \varepsilon$$

Then we have

$$A \subseteq \bigcup_{k=1}^{\infty}[a_k, b_k], \quad \sum_{k=1}^{\infty}(f(b_k) - f(a_k)) < \lambda_f^*(A) + \varepsilon$$

Thus

$$\Lambda_f^*(A) \leq \lambda_f^*(A) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\Lambda_f^*(A) \leq \lambda_f^*(A)$$

2. Let $[a_1, b_1], [a_2, b_2], \cdots$ be a sequence of closed intervals , such that

$$B \subseteq \bigcup_{k=1}^{\infty} [a_k, b_k], \quad \sum_{k=1}^{\infty} (f(b_k) - f(a_k)) < \Lambda_f^*(B) + \varepsilon$$

If f is left-continuous, for each $k \in \mathbb{N}$, there is a'_k, such that

$$f(a_k) - f(a'_k) < \frac{\varepsilon}{2^k}$$

Then

$$B \subseteq \bigcup_{k=1}^{\infty} (a'_k, b_k], \quad \sum_{k=1}^{\infty} \left(f(b_k) - f(a'_k)\right) < \Lambda_f^*(B) + 2\varepsilon$$

That is

$$\lambda_f^*(B) \le \Lambda_f^*(B) + 2\varepsilon$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lambda_f^*(B) \le \Lambda_f^*(B)$$

□

1.7 Hausdorff Measure

Definition 1.7.1 ▸ Hausdorff Outer Measure

Hausdorff measure is defined in terms of an auxiliary set function that we introduce first. Let $0 \le s < \infty$, $0 < \varepsilon \le \infty$, and let $A \subset \mathbb{R}^n$. Define

$$H_\varepsilon^s(A) = \inf\left\{\sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam} E_i)^s : A \subset \bigcup_{i=1}^{\infty} E_i, \operatorname{diam} E_i < \varepsilon\right\},$$

where $\alpha(s)$ is a normalization constant defined by

$$\alpha(s) = \frac{\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2} + 1)},$$

with

$$\Gamma(t) = \int_0^\infty e^{-x} x^{t-1}\, dx, \quad 0 < t < \infty.$$

It follows from the definition that if $\varepsilon_1 < \varepsilon_2$, then $H^s_{\varepsilon_1}(E) \geq H^s_{\varepsilon_2}(E)$. This allows the following, which is the definition of ***s-dimensional Hausdorff measure***:

$$H^s(A) = \lim_{\varepsilon \to 0} H^s_\varepsilon(A) = \sup_{\varepsilon > 0} H^s_\varepsilon(A).$$

Remark.
- Hausdorff Measure can be defined in any metric space.
- The sets E_i in the definition of $H^s_\varepsilon(A)$ are arbitrary subsets of $\mathbb{R}^n$. However, they could be taken to be closed sets, since $\operatorname{diam} E_i = \operatorname{diam} \overline{E_i}$.

Theorem 1.7.2

For each nonnegative number s, H^s is a Carathéodory outer measure.

Proof. First we show that H^s is an outer measure. We only show that H^s is subadditive, the other properties are obvious. Let $A_1, A_2, \cdots$ be a sequence of sets with H^s- outer measure finite. Let $A = \bigcup_{k=1}^\infty A_k$. For any $\varepsilon > 0$, and for each k, there are sets $E_1^{(k)}, \cdots$ such that

$$A_k \subseteq \bigcup_{i=1}^\infty E_i^{(k)}, \quad \sum_{i=1}^\infty \alpha(s)\, 2^{-s} \left(\operatorname{diam} E_i^{(k)}\right)^s < H^s_\varepsilon(A_k) + \frac{\varepsilon}{2^k}$$

We can arange $\left\{E_i^{(k)}\right\}_{i,k \in \mathbb{Z}_{>0}}$ to be a sequence of sets such that

$$A \subseteq \bigcup_{i,k \in \mathbb{Z}_{>0}} E_i^{(k)}, \quad \sum_{i,k \in \mathbb{Z}_{>0}} \alpha(s)\, 2^{-s} \left(\operatorname{diam} E_i^{(k)}\right)^s < \sum_{k=1}^\infty H^s_\varepsilon(A_k) + \varepsilon$$

Thus

$$H^s_\varepsilon(A) \leq \sum_{k=1}^\infty H^s_\varepsilon(A_k) + \varepsilon$$

Since $H_\varepsilon^s(A_k) \le H^s(A_k)$ for each k, we have

$$H_\varepsilon^s(A) \le \sum_{k=1}^{\infty} H^s(A_k) + \varepsilon$$

Let $\varepsilon > 0$, we have

$$H^s(A) \le \sum_{k=1}^{\infty} H^s(A_k)$$

Now we show that H^s is a Carathéodory outer measure. Suppose that A, B be two sets with $\mathbf{d}(A,B) > 0$, and suppose $0 < \varepsilon < \mathbf{d}(A,B)$. Let $\{E_k\}$ be a covering of $A \cup B$ with $\operatorname{diam}(E_i) < \varepsilon$ .Then each E_k cannot intersect with both A and B. Let $\mathcal{A}$ be the collection of E_k intersecting with A , and Let $\mathcal{B}$ be that of intersecting with B. Then

$$\begin{aligned}\sum_{k=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam} E_i)^s &= \sum_{E_k \in \mathcal{A}} \alpha(s) 2^{-s} (\operatorname{diam} E_k)^s + \sum_{E_k \in \mathcal{B}} \alpha(s) 2^{-s} (\operatorname{diam} E_k)^s \\ &\ge H_\varepsilon^s(A) + H_\varepsilon^s(B)\end{aligned}$$

We can take $\{E_i\}$ such that

$$\sum_{k=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam} E_i)^s < H_\varepsilon^s(A \cup B) + \varepsilon$$

Thus we have

$$H_\varepsilon^s(A \cup B) \ge H_\varepsilon^s(A) + H_\varepsilon^s(B) + \varepsilon$$

Since $0 < \varepsilon < \mathbf{d}(A,B)$ is arbitrary, take $\varepsilon \to 0^+$, it follows that

$$H^s(A \cup B) \ge H^s(A) + H^s(B)$$

, which shows that H^s is a Carathéodory outer measure. □

Theorem 1.7.3

For each $A \subset \mathbb{R}^n$, there exists a Borel set $B \supset A$ such that

$$H^s(B) = H^s(A).$$

Proof. Let $\{\varepsilon_j\}$ be a decreasing suquence of positive numbers such that $\varepsilon_j \to 0$ as $j \to \infty$. For each j, there is a suquence of closed set (Remark of Definition

1.7.1) $E_{i,j}$, such that

$$B \subseteq \bigcup_{i=1}^{\infty} E_{i,j}, \quad \sum_{i=1}^{\infty} \alpha(s) 2^{-s} \left(\operatorname{diam}\left(E_{i,j}\right)\right)^{s} < H_{\varepsilon_j}^{s}(B) + \varepsilon_j$$

Let $A_j := \bigcup_{i=1}^{\infty} E_{i,j}$, and let $A := \bigcap_{j=1}^{\infty} A_j$. Then A is a Borel set such that $B \subseteq A$. Note that

$$H_{\varepsilon_j}^{s}(B) + \varepsilon_j > \sum_{i=1}^{\infty} \alpha(s) 2^{-s} \left(\operatorname{diam}\left(E_{i,j}\right)\right)^{s} \geq H_{\varepsilon_j}^{s}\left(A_j\right) \geq H_{\varepsilon_j}^{s}(A)$$

Let $j \to \infty$, we have

$$H^{s}(B) \geq H^{s}(A) \implies H^{s}(B) = H^{s}(A)$$

$\square$

Theorem 1.7.4

Suppose $A \subset \mathbb{R}^n$ and $0 \leq s < t < \infty$. Then
(i) If $H^s(A) < \infty$ then $H^t(A) = 0$.
(ii) If $H^t(A) > 0$ then $H^s(A) = \infty$.

Proof. (ii) is just a restatement of (i). We now show that (i) holds. Take any $\varepsilon > 0$, there are $\{E_i\}$ with $\operatorname{diam}(E_i) \leq \varepsilon$ such that

$$A \subseteq \bigcup_{i=1}^{\infty} E_i, \quad \sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam}(E_i))^{s} < H_{\varepsilon}^{s}(A) + \varepsilon \leq H^{s}(A) + \varepsilon$$

Then

$$\begin{aligned} H_{\varepsilon}^{t}(A) &\leq \sum_{i=1}^{\infty} \alpha(t) 2^{-t} (\operatorname{diam}(E_i))^{t} \\ &= \frac{\alpha(t)}{\alpha(s)} 2^{s-t} \sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam}(E_i))^{s} (\operatorname{diam}(E_i))^{t-s} \\ &\leq \frac{\alpha(s)}{\alpha(t)} 2^{s-t} \varepsilon^{t-s} \left(H^{s}(A) + \varepsilon\right) \end{aligned}$$

Let $\varepsilon \to 0$, we have

$$H^t(A) \le 0 \implies H^t(A) = 0$$

$\square$

Exercises for Hausdorff Measure

Exercise 1.7.5

If $A \subset \mathbb{R}$ is an arbitrary set, show that $H^1(A) = \lambda^*(A)$.

Remark.

In this problem, you will see the importance of the constant that appears in the definition of Hausdorff measure. The constant $\alpha(s)$ that appears in the definition of H^s-measure is equal to 2 when $s = 1$. That is, $\alpha(1) = 2$, and therefore, the definition of $H^1(A)$ can be written as

$$H^1(A) = \lim_{\varepsilon \to 0} H^1_\varepsilon(A),$$

where

$$H^1_\varepsilon(A) = \inf\left\{\sum_{i=1}^\infty \operatorname{diam} E_i \,:\, A \subset \bigcup_{i=1}^\infty E_i \subset \mathbb{R}, \operatorname{diam} E_i < \varepsilon\right\}.$$

This result is also true in $\mathbb{R}^n$ but is more difficult to prove. The isodiametric inequality $\lambda^*(A) \le \alpha(n)\left(\frac{\operatorname{diam} A}{2}\right)^n$ (whose proof is omitted in this book) can be used to prove that $H^n(A) = \lambda^*(A)$ for all $A \subset \mathbb{R}^n$.

Proof. For any $\varepsilon > 0$, there are closed intervals $I_1, \cdots$ with $v(I_i) < \varepsilon$ such that

$$A \subseteq \bigcup_{i=1}^\infty I_i, \quad \sum_{i=1}^\infty v(I_i) < \lambda^*(A) + \varepsilon$$

Then

$$\sum_{i=1}^\infty \alpha(1)\, 2^{-1} (\operatorname{diam} I_i)^1 = \sum_{i=1}^\infty v(I_i) < \lambda^*(A) + \varepsilon$$

Thus

$$H^1_\varepsilon(A) < \lambda^*(A) + \varepsilon$$

Take $\varepsilon \to 0$, we have

$$H^1(A) \le \lambda^*(A)$$

The other hand, take $\varepsilon > 0$, there are closed set E_i with $\operatorname{diam} E_i < \varepsilon$, such that

$$A \subseteq \bigcup_{i=1}^{\infty} E_i, \quad \sum_{i=1}^{\infty} \operatorname{diam} E_i < H_\varepsilon^1(A) + \varepsilon$$

Note that

$$\operatorname{diam} E_i = \sup E_i - \inf E_i$$

Let

$$I_i := \left[\inf E_i - \frac{\varepsilon}{2^{i+1}}, \sup E_i + \frac{\varepsilon}{2^{i+1}}\right]$$

Then

$$E_i \subseteq I_i, \quad v(I_i) < \operatorname{diam} E_i + \frac{\varepsilon}{2^i}$$

Thus

$$A \subseteq \bigcup_{i=1}^{\infty} I_i, \quad \sum_{i=1}^{\infty} v(I_i) < H_\varepsilon^1(A) + 2\varepsilon < H^1(A) + 2\varepsilon$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lambda^*(A) \le H^1(A)$$

□

Exercise 1.7.6

For $A \subset \mathbb{R}^n$, use the isodiametric inequality introduced in Exercise 4.1 to show that

$$\lambda^*(A) \le H^n(A) \le \alpha(n)\left(\frac{\sqrt{n}}{2}\right)^n \lambda^*(A).$$

Proof. For each $\varepsilon > 0$, there are sets $E_1, E_2, \cdots$ with $\operatorname{diam} E_i < \varepsilon$ such that

$$A \subseteq \bigcup_{i=1}^{\infty} E_i, \quad \sum_{i=1}^{\infty} \alpha(n)\, 2^{-n} (\operatorname{diam}(E_i))^n < H_\varepsilon^n(A) + \varepsilon$$

Then

$$\lambda^*(A) \leq \sum_{i=1}^{\infty} \lambda^*(E_i) \leq \sum_{i=1}^{\infty} \alpha(n) 2^{-n} (\operatorname{diam}(E_i))^n < H_\varepsilon^n(A) + \varepsilon$$

Take $\varepsilon \to 0$, then

$$\lambda^*(A) \leq H^n(A)$$

For each $\varepsilon > 0$, there are closed intervals $I_1, I_2, \cdots$ such that

$$A \subseteq \bigcup_{i=1}^{\infty} I_i, \quad \sum_{i=1}^{\infty} v(I_i) < \lambda^*(A) + \varepsilon$$

Since for each I_i, there are finitely may squares $J_{i,j}$, such that

$$I_i \subseteq \bigcup_{j=1}^{\infty} J_{i,j}, \quad \sum_{j=1}^{\infty} v\left(J_{i,j}\right) < v(I_i) + \frac{\varepsilon}{2^i}$$

We can arange $\left\{J_{i,j}\right\}_{i,j \in \mathbb{Z}_{>0}}$ into a sequence of squares $\{K_m\}$ such that

$$A \subseteq \bigcup_{m=1}^{\infty} K_m, \quad \sum_{i=1}^{\infty} v(K_m) < \lambda^*(A) + 2\varepsilon$$

Note that

$$\left(\frac{1}{\sqrt{n}} \operatorname{diam}(K_m)\right)^n = v(K_m)$$

We have

$$\begin{aligned} H_\varepsilon^n(A) \leq \sum_{i=1}^{\infty} \alpha(n) 2^{-n} (\operatorname{diam}(K_m))^n &= \alpha(n) \left(\frac{\sqrt{n}}{2}\right)^n \sum_{i=1}^{\infty} v(K_m) \\ &\leq \alpha(n) \left(\frac{\sqrt{n}}{2}\right)^n (\lambda^*(A) + 2\varepsilon) \end{aligned}$$

Take $\varepsilon \to 0$, we have

$$H^n(A) \le \alpha(n) \left(\frac{\sqrt{n}}{2}\right)^n \lambda^*(A)$$

□

Exercise 1.7.7

Show that $H^s \equiv 0$ on $\mathbb{R}^n$ for all $s > n$.

Proof. Let

$$I_k := [-k, k]^n$$

For any $A \subseteq \mathbb{R}^n$, we have

$$A \subseteq \bigcup_{k=1}^{\infty} I_k$$

Isodiametric inequality gives that

$$H^n(I_k) \le \alpha(n) \left(\frac{\sqrt{n}}{2}\right)^n \lambda^*(I_k) < \infty$$

Then Theorem 1.7.4 gives that

$$H^s(I_k) = 0$$

Thus we have

$$H^s(A) \le \sum_{k=1}^{\infty} H^s(I_k) = 0$$

□

Exercise 1.7.8

Let $C \subset \mathbb{R}^2$ denote the circle of radius 1 and regard C as a topological space with the topology induced from $\mathbb{R}^2$. Define an outer measure $\mathcal{H}$ on C by

$$\mathcal{H}(A) := \frac{1}{2\pi} H^1(A) \quad \text{for any set } A \subset C.$$

Later in the book, we will prove that $H^1(C) = 2\pi$. Thus, you may assume that. Show that

(i) $\mathcal{H}(C) = 1$.

(ii) Prove that $\mathcal{H}$ is a Borel regular outer measure.

(iii) Prove that $\mathcal{H}$ is rotationally invariant; that is, prove that for every set $A \subset C$, one has $\mathcal{H}(A) = \mathcal{H}(A')$, where A' is obtained by rotating A through an arbitrary angle.

(iv) Prove that $\mathcal{H}$ is the only outer measure defined on C that satisfies the previous three conditions.

1.8 Hausdorff Dimension of Cantor Sets

Definition 1.8.1 ▸ General Cantor set

Let $0 < \gamma < 1/2$ and set $I_{0,1} = [0, 1]$. Let $I_{1,1}$ and $I_{1,2}$ denote the intervals $[0, \gamma]$ and $[1 - \gamma, 1]$ respectively. They result by deleting the open middle interval of length $1 - 2\gamma$. At the next stage, delete the open middle interval of length $\gamma(1 - 2\gamma)$ of each of the intervals $I_{1,1}$ and $I_{1,2}$. There remain 2^2 closed intervals each of length γ^2. Continuing this process, at the kth stage there are 2^k closed intervals each of length γ^k. Denote these intervals by $I_{k,1}, \dots, I_{k,2^k}$. We define the generalized Cantor set as

$$C(\gamma) = \bigcap_{k=0}^{\infty} \bigcup_{j=1}^{2^k} I_{k,j}.$$